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Abstract

Atherosclerosis is a lipid-driven inflammatory disease affecting more than 500 million people yearly. It is characterized by plaque formation within the arteries that can lead to blocked arteries and strokes. Risk factors include a high concentration of LDL and a low concentration of HDL. Additionally, excess LDL is prone to oxidation in the arteries, which can attract monocytes that differentiate into macrophages that can either be atheroprotective or atherogenic, depending on their phenotype. Currently, there are suggested to be six distinct macrophage subtypes: resident-like macrophages, inflammatory macrophages, LAMs, iLAMs, IFN γ macrophages, and proliferating macrophages. To study macrophage heterogeneity, mouse models are used due to the ability to study atherosclerosis in a controlled environment and reduce costs. Among different genetically modified mouse models, ApoE $^{-/-}$ and LDLR $^{-/-}$ are primarily used to study macrophage heterogeneity. Tissue samples from these models can be analyzed using scRNA-seq. However, not all findings using mouse models are translatable to humans. The aim of this study was to analyze and compare transcriptional differences between atherosclerotic mouse and human macrophages to evaluate the translational value of using mouse as a model organism in human atherosclerosis research. To achieve this, atherosclerotic plaque samples were collected and analyzed using a well-established scRNA-seq workflow. Analysis revealed that the resident-like macrophages, inflammatory macrophages, and LAMs were largely conserved across species. In contrast, the iLAMs, and IFN γ macrophages showed several species-specific differentially expressed genes. Additionally, the proliferating macrophages were not found in the mouse macrophage populations. Despite several methodological limitations, these findings suggest that mouse models show promising translatability of macrophage findings to humans in atherosclerosis research.

Contents

1	Introduction	3
2	Methods	4
2.1	Single-cell data	4
2.2	Uniform processing workflow	4
2.3	Subsetting myeloid cells and annotation	5
2.4	Subsetting macrophages and annotating subtypes	5
2.5	Downstream analysis	5
2.6	Data and code availability	6
3	Results	7
3.1	Macrophages dominate the myeloid cell landscape of atherosclerotic plaques	7
3.2	Atherosclerotic plaques in mouse and human show distinct distributions of macrophage subsets	8
3.3	Evaluation of macrophage subtype distributions across species	9
3.4	Comparative analysis of differentially expressed genes	10
3.4.1	Species-specific marker genes of resident-like macrophages in mouse and human	11
3.4.2	Inflammatory macrophages are largely conserved across species	13
3.4.3	Comparison of differentially expressed genes between mouse and human LAMs	14
3.4.4	Absence of <i>TREM1</i> expression in mouse iLAMs compared to human iLAMs	14
3.4.5	IFNIC macrophage populations exhibit mouse-specific genes	15
4	Discussion	16
5	Acknowledgments	17
6	Supplementary material	20
6.1	Gene expression omnibus of available datasets	20
6.2	Cluster plots	22
6.2.1	Cluster plots of all mouse immune cells	22
6.2.2	Cluster plots of mouse myeloid subset	24
6.2.3	Cluster plots of mouse macrophage subset	25
6.2.4	Cluster plots of all human immune cells	26
6.2.5	Cluster plots of human myeloid subset	28
6.2.6	Cluster plots of human macrophage subset	29
6.3	Annotation of macrophage subtypes	30
6.4	Transcription factor analysis	30
6.5	Raw result of mouse and human differential expression analysis	32
6.5.1	Mouse macrophage gene markers	32
6.5.2	Human macrophage gene markers	35

1 Introduction

Atherosclerosis is a lipid-driven inflammatory disease, affecting more than 500 million people yearly [1]. It is characterized by plaque formation that can lead to blocked arteries, resulting in possible symptoms such as ruptures, strokes, and other severe complications [2]. Important risk factors for plaque formation and progression of atherosclerosis include a high concentration of low-density lipoproteins (LDLs), and a low concentration of high-density lipoproteins (HDLs) [3]. LDL particles transport cholesterol (LDL-c) from the liver to peripheral tissues, leading to plaque build-up in the arteries. HDL particles on the other hand transport cholesterol (HDL-c) from peripheral tissues to the liver for breakdown [4, 5]. To study atherosclerosis, genetically modified mice are used to increase translatability of findings in mouse models to humans. Among different genetically modified mouse models, low-density lipoprotein receptor (LDLR)^{-/-} and apolipoprotein E (apoE)^{-/-} mice are widely utilized, since the respective knockouts lead to changes in lipid metabolism resulting in increased cholesterol levels. LDL receptors are responsible for cholesterol homeostasis and the uptake of LDL through receptor-mediated endocytosis, leading to less cholesterol accumulation [6]. Consequently, a LDLR^{-/-} mouse has increased cholesterol levels compared to the wild-type counterpart. Additionally, apoE is an apolipoprotein that acts as a ligand for LDL receptors, thereby contributing to lipid metabolism [7]. Each genetically modified model has its own advantages and disadvantages. Whereas LDLR^{-/-} mice are used to study lipid metabolism, apoE^{-/-} mice are used to study inflammation and lesions within atherosclerosis [3].

Excess LDL in the arteries is prone to oxidation, which will attract monocytes that differentiate into macrophages. These macrophages will subsequently scavenge the oxidized LDL. However, excess scavenging of oxidized LDL can turn macrophages into lipid-laden macrophages that promote disease progression. Additionally, other macrophage subtypes also play a role in disease progression. Currently, there are suggested to be six macrophage subtypes: inflammatory macrophages, resident-like macrophages, lipid-associated macrophages (LAMs), inflammatory LAMs (iLAMs), IFN γ macrophages, and proliferating macrophages [8, 9]. Inflammatory macrophages are responsible for lesional inflammation and show atherogenic characteristics, whereas resident-like macrophages and LAMs show atheroprotective characteristics and are involved in endocytosis and cholesterol metabolism, respectively [10]. Dib et al. (2023) [9] recently identified a novel macrophage subtype, iLAMs. Unlike LAMs, iLAMs primarily show atherogenic characteristics and have been shown to induce lipid-driven inflammation within atherosclerotic plaques. However, the roles of IFN γ macrophages and proliferating macrophages in atherosclerosis have yet to be elucidated.

To gain insight into macrophage heterogeneity in atherosclerosis, single-cell RNA sequencing (scRNA-seq), which is a technique that isolates individual cells, extracts their RNA, and generates a transcriptional profile, has been used to provide a detailed description of macrophage subtypes. Many of these scRNA-seq studies utilize data from atherosclerotic plaques of genetically modified mice. While findings using mouse models in combination with scRNA-seq have contributed to acquiring insights into macrophage heterogeneity, not all findings are translatable to human atherosclerosis. For example, the study by Horstmann et al. (2024), reported that transcription of immune cells of mouse and human did not match, challenging the suitability of using mouse models to gain insights into human atherosclerosis, especially for the adaptive immune system [11]. However, another study by Zerneck et al. (2023), reported that mouse and human exhibited similar transcriptional profiles in mononuclear phagocytes [12], supporting the applicability of using mouse models to study human atherosclerosis with regards to the function of mononuclear phagocytes. The findings by Horstmann et al. (2024) [11] and Zerneck et al. (2023) [12] suggest that findings of certain immune cell populations might be more translatable to humans compared to other immune cell populations.

Therefore, this research aims to analyze transcriptional differences within macrophage and macrophage subtypes between atherosclerotic mouse and human, and to provide insights into the suitability of using mouse models to study human atherosclerosis by including more human data compared to previous research. To achieve this, single-cell RNA sequencing (scRNA-seq) data of mouse carotid plaques and human aortic plaques (N = 36) have been compiled and analyzed. The data were subsequently automatically annotated, and the myeloid cells were subset for downstream analysis. Afterwards, the macrophages in the myeloid cell population were manually identified and subset for downstream analysis. The macrophage subtypes were manually identified and gene expression analysis was performed for each sample and organism to find differentially expressed genes. Lastly, the results of a differential gene expression (DGE) analysis were compared across species, to find genes that are strongly upregulated in one population compared to all the other populations.

2 Methods

A well-established scRNA-seq processing workflow using Seurat V5 [13] was applied to selected scRNA-seq datasets. In total, 9 integrative analyses were performed. These included a separate analysis for each study and for each species. In the species-level integration, the mouse samples were merged by condition, instead of by sample. This approach was necessary due the low number of macrophage cells which would cause issues downstream.

2.1 Single-cell data

Seven atherosclerotic mouse aortic plaque datasets, obtained from three studies, and 36 atherosclerotic human carotid plaque samples, obtained from five studies [9, 11, 14–16]. Mouse datasets were included if they were ApoE^{-/-} or LDLR^{-/-}, used a live CD45+ sorting strategy, and were prepared using 10x Genomics Single Cell 3' reagent kit v2 (Table 1).

Study	Model	Diet	Age (weeks)
Cochain et al. (2018) [17]	LDLR ^{-/-}	Chow	20
Cochain et al. (2018) [17]	LDLR ^{-/-}	High fat	11
Cochain et al. (2018) [17]	LDLR ^{-/-}	High fat	11
Vafadarnejad et al. (2020) [18]	LDLR ^{-/-}	High fat	10
Winkels et al. (2018) [19]	ApoE ^{-/-}	Chow	8
Winkels et al. (2018) [19]	ApoE ^{-/-}	Chow	20
Winkels et al. (2018) [19]	ApoE ^{-/-}	Western	20 (started with Western diet at 8 weeks)

Table 1: Overview of the mouse data used.

2.2 Uniform processing workflow

Using the guidelines provided by [20], [21], a scRNA-seq processing workflow was designed for each study, species and both species combined. Firstly, doublets were removed with scDbtFinder to ensure that only data of singlets is captured [22]. Subsequently quality control (QC) was performed to remove low-quality cells. This was done by filtering out cells that expressed less than 200 genes or contained a high mitochondrial content per cell. A limit for the mitochondrial content per cell was set because a high mitochondrial content is associated with dying cells. For mouse datasets, this limit was set at 5%, while the limit for human datasets was set at 10%. Secondly, the data was log-normalized to ensure comparability by reducing sampling effects. Afterwards, the “FindVariableFeature” function of Seurat V5 was used with the VST selection method and the data was scaled by regressing out the percentage of mitochondrial content and the number of genes expressed per cell. Subsequently, a principal component analysis (PCA) was performed to find the most informative principal components (PCs) and a JackStraw score was calculated to determine the statistical significance of individual PCs. The Jackstraw score was then used to find the number of significant ($p < 0.05$) PCs, which were used as the number of dimensions defined in a Uniform Manifold Approximation and Projection (UMAP), using PCA for the reduction argument. Lastly, a CCA-based integration was performed over each condition in the mouse datasets and each sample in the human datasets to remove batch effects. Integration was done by using Seurat V5’s “IntegrateLayers” function.

After integration, another PCA was performed and a Jackstraw score was calculated to find the significant PCs of the integrated data. The number of significant PCs was used as the dimension parameter of Seurat’s “Find-

Neighbors” function, which constructs a k-nearest neighbors graph. Following this, the “FindClusters” function was used to identify clusters within the constructed graph. Additionally, several different resolutions were used as an argument in the “FindClusters” function to find a stable cluster assignment. To compare the stability of these cluster assignments, the “clustree” R package was used [23]. Lastly, another UMAP was performed, this time using the CCA integration for the reduction argument.

To visualize the integration results, an integrated UMAP analysis was performed using the significant PCs as the dimension parameter. The data were annotated using the R packages SingleR [24] and celltex’s “BluePrint-Encode” as a reference dataset for the human datasets, and “MouseRNASeq” for the mouse datasets.

2.3 Subsetting myeloid cells and annotation

After the uniform preprocessing, clustering and annotation with SingleR described in the previous section, a cluster-based selection was performed where all identified myeloid cells were taken of clusters consisting of at least 50% myeloid cells. It was assumed that if only a small percentage of cells in a cluster were classified as myeloid, then these annotations were likely misclassified and therefore excluded. The threshold was set because if a high proportion of cells in that cluster were classified as myeloid, then it is likely that these annotations were not misclassifications. The count data of this subset were preprocessed again, following the workflow described in Section 2.2. However, after taking the subset of myeloid cells, some samples contained few cells. The low amount of cells would cause a problem when using “IntegrateLayers”. The “IntegrateLayers” function works by finding subset of cells from one integration batch and ‘anchoring’ them to similar subsets of another integration batch. The robustness of these anchors are based on two components, the anchor score and a weight. In the calculation of the weight, the gaussian kernel is applied to the distance between each cell and its anchor. This value is subsequently normalized by summing over all anchors, where the number of normalizing anchors is defined as `k.weights` and is defaulted at 100. However, because some samples had fewer than 100 cells, this would cause the “IntegrateLayers” to not be able to normalize over 100 anchors, since the amount of anchors is less than or equal to the amount of cells. To handle this, we decided to only include samples that had more than 50 cells, and set the `k.weight` parameter equal to the amount of cells present in the sample with the lowest cell count. This would result in a possible under-correction of batch effects. However, lowering the amount of anchors used in the normalization, causes a lower weight value for each query cell and their anchor, leading to less correction [25].

After integration, DGE analysis was performed using the “FindAllMarkers” function with default parameters. Additionally, the Human Protein Atlas (HPA) [26], evidence from literature and markers from the Panglao database [27] were used to identify cell types. The HPA offers information on the normalized transcripts per million (nTPM) in immune cell types value for genes. This information was used to identify the cell type in which a specific gene had the highest normalized transcripts per million (nTPM) value.

2.4 Subsetting macrophages and annotating subtypes

The clusters identified as macrophages were then subset and the corresponding count data were preprocessed. A DGE analysis was then performed on each cluster (Table S2, Table S3), and the differentially expressed genes were subsequently used for manual annotation, similarly to the method described in Section 2.3. In contrast to our manual annotation of the myeloid cells, the macrophage subsets were annotated by also using marker genes from [8, 10]. In addition, dot plots and violin plots of these marker genes were utilized (Table 2). Mouse orthologs of the human marker genes were found using the NCBI Gene database [28]. Based on the expression of marker genes per cluster and evidence from literature, a macrophage subtype was assigned to each cell. Lastly, the resulting annotations were visualized in a UMAP plot.

2.5 Downstream analysis

To analyze the macrophage subtypes in each species, DGE analysis was performed using Seurat’s “FindAllMarkers” function [13], which is a technique that finds genes that are significantly upregulated in one cluster compared against all other clusters by using a Wilcoxon signed-rank test. Moreover, only genes that are up-regulated by a fold change of 0.25 and expressed in at least 25% of a cluster were included. Additionally, the

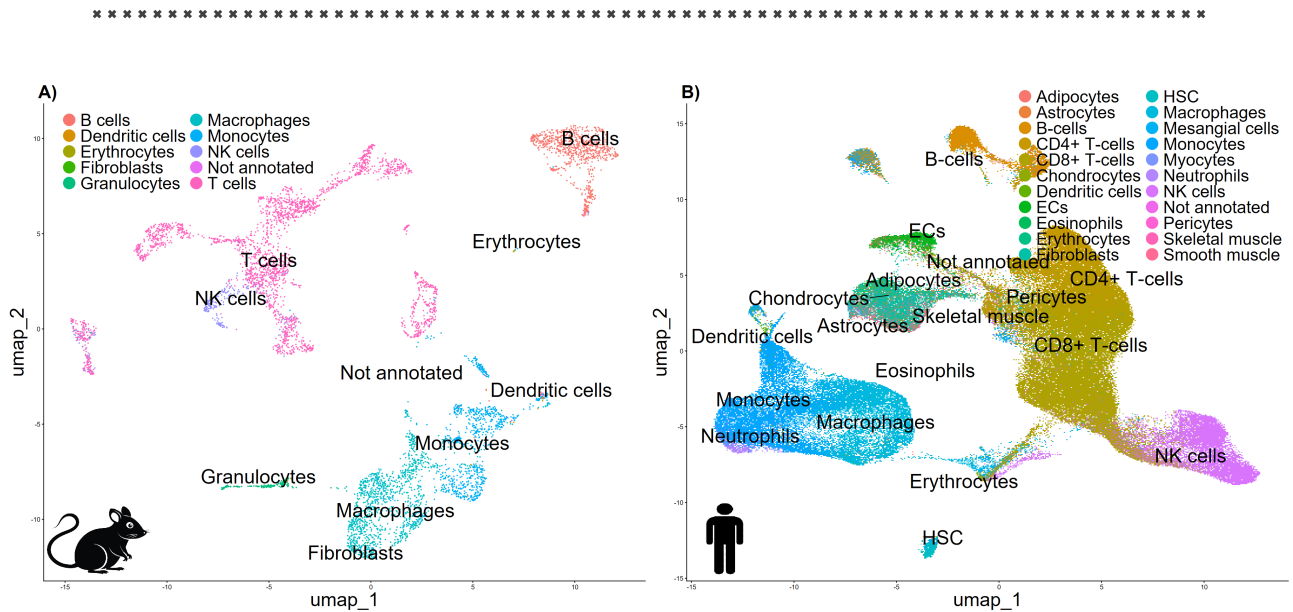
Macrophage subtype	Function	Marker genes
Resident-like macrophages	Endocytosis	LYVE1, FOLR2, C1QA, CX3CR1, SELENOP
Inflammatory macrophages	Inflammation	S100A8, S100A9, NLRP3, IL1B, CXCL2
Lipid associated macrophages	Lipid metabolism	TREM2, FABP5, APOE, APOC1
Inflammatory lipid associated macrophages	Inflammation and lipid metabolism	TREM1, PLIN2, CCL2, CCL7, VCAN
IFN γ macrophages	Disease progression	ISG15, IRF7, IFIT3, IFIT1
Proliferating macrophages	Proliferation	STMN1, TOP2A, MKI67, TUBA1B

Table 2: Overview of macrophage subtypes, the function, and corresponding marker genes of each macrophage subtype. Information on the function of the macrophage subtypes and marker genes were obtained from the study by Dib et al. (2023), and the review papers of Winther et al. (2022) and Willemsen et al. (2022) [8–10]

relative frequencies between cell type annotations were also evaluated to find systemic differences across the macrophage landscape, and a transcription factor (TF) analysis using decoupleR was conducted [29]. Lastly, another DGE analysis was performed on each subtype, taking into account all differentially expressed genes. DGE analysis across species were then compared by merging DGE results based on Entrez IDs of homologs. To achieve this, human genes were first converted to corresponding Entrez IDs by using a mapping file acquired from SynGO [30]. Additionally, mouse genes were converted to the Entrez ID of corresponding human homologs by using the R package “homologene” [31]. Finally, both DGE analysis results were merged based on human Entrez IDs. This approach resulted in the best performance compared to solely using SynGO or homologene.

2.6 Data and code availability

Most of the scRNA-seq datasets used are publicly available and can be found by their respective Gene Expression Omnibus GSE identifier (Table S1). Only the scRNA-seq data from Depuydt et al. (2023) [15] is not publicly available. Data and scripts developed in this study can be found on https://github.com/philip-ahm/macrophage_sc_analysis. The file structure follows the ENCORE guidelines [32]. Additionally, the project environment will be captured using “renv” [33] to ensure reproducibility.



3 Results

The result section first evaluates the results of the automatic annotation of broad immune cells, manual annotations of myeloid cells, which include the dendritic cells (DCs), macrophages and monocytes, and the manual annotation of macrophage subtypes. Additionally, the results of DGE analysis are evaluated.

3.1 Macrophages dominate the myeloid cell landscape of atherosclerotic plaques

After QC there were 6,172 mouse immune cells, and 110,319 human immune cells. Automatic annotation revealed that mouse plaques consisted of 36% myeloid cells whereas human plaques consisted of 23.2% myeloid cells. The automatic annotations were visualized in a UMAP plot (Figure 1). Following automatic annotation, a subset of myeloid cells were taken, as described in Section 2.3, resulting in a reduction of myeloid cells of mouse plaques (−0.9%) and human plaques (−1.9%). In total, the mouse plaques contained 2,169 myeloid cells, whereas the human plaques contained a total of 23,541 myeloid cells. This subset was then analyzed as described in Section 2.3.

For manual annotation of the mouse myeloid cells, a resolution of 0.2 was used for the low-resolution analysis, revealing 7 distinct clusters, and a resolution of 1.2 was used for the high-resolution analysis, revealing 16 distinct clusters (Figure S2). A high-resolution analysis was performed to ensure identification of macrophage populations that show subtle differences compared to monocytes. Afterward, DGE analysis was performed on each cluster to find marker genes. These marker genes were then used to identify the cell type of a cluster. The clusters corresponding to the same annotation were then merged (Figure 2A). Another DGE analysis was performed to identify cell types, revealing that the macrophage population differentially expressed macrophage gene markers such as *C1qa/b/c*, *Folr2*, and *Maf* [10] (Figure 2B). The DC population differentially expressed *Cd209a*, *H2-Oa*, and *Siglecg*, which are associated with DCs [34], suggesting correct annotation for this cell type. Furthermore, the monocyte population also expressed known monocyte markers such as *Cd177*, *Ly6c2*, and *Vcan* [34]. Lastly, a T cell cluster was identified that expressed T cell markers *Cd8b1*, *Lck*, and *Rag1* [8].

For the manual annotation of the human myeloid cells, two resolutions were used to find clusters within the myeloid cell subset: a low resolution of 0.3 and a high resolution of 1.5. Low-resolution clustering revealed 10 distinct clusters, whereas high-resolution clustering revealed 28 distinct clusters (Figure S5). Similarly to the annotation of human myeloid cells, a DGE analysis was performed. This revealed that the macrophage population exhibited strong expression of known macrophage marker genes *SELENOP*, *APOE* and *APOC1* [10], the monocyte population strongly expressed monocyte markers *RIPOR2*, *FCN1*, and *APOBEC3A* [26], and the DC population expressed DC markers such as *FCER1A*, *CD1C* and *FLT3* [9, 26]. Additionally, a natural killer

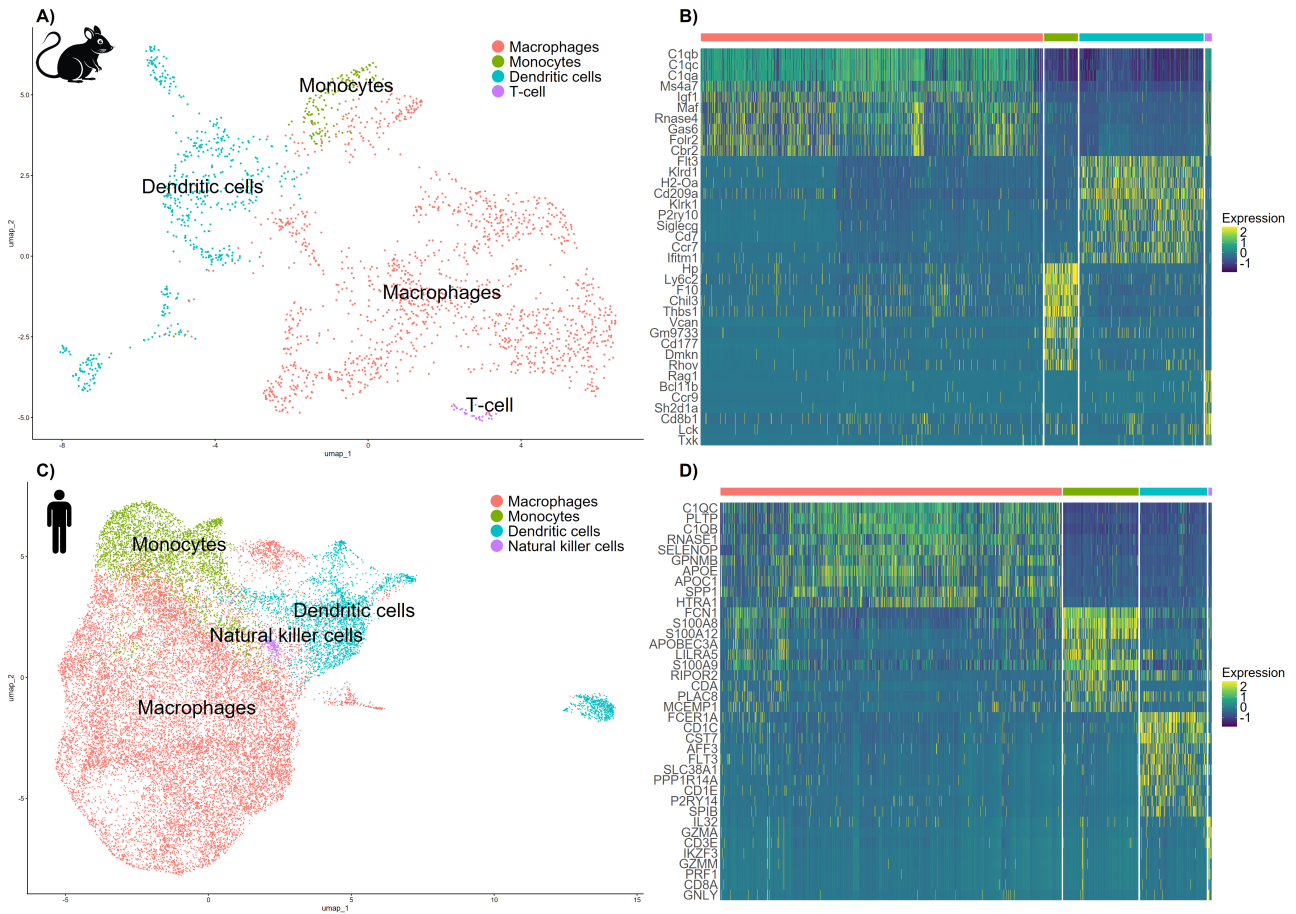


Figure 2: Results of annotation and the heatmap of the differentially expressed genes of each identified cell type. (A) UMAP visualization of the final annotations of each cluster in mouse. (B) Heatmap of differentially expressed genes with the highest log-fold change in the annotated clusters in mouse. The colored bar on top of the heatmap represents the cell type, see the legend of panel A. (C) UMAP visualization of the final annotations of each cluster in human. (D) Heatmap of differentially expressed genes with the highest log-fold change in the annotated clusters in mouse. The colored bar on top of the heatmap represents the cell type, see the legend of panel C.

(NK) cell cluster was identified, which showed strong expression of NK marker genes *GZMA*, *GNLY*, and *PRF1* [8].

Evaluation of the distributions in myeloid cells showed that mouse myeloid cell landscape consisted of 67.5% macrophages, 6.6% monocytes, 24.5% DCs, and 1.3% T cells, suggesting that the SingleR annotation misclassified some T-cells as myeloid. In the human dataset, myeloid cells consisted of 70% macrophages, 15.5% monocytes, 13.8% DCs, and 0.8% natural killer cells, which was likely also a misclassification made by SingleR (Table 3). In total, 1,464 macrophages were identified in the mouse data, whereas the human data consisted of 16,349 macrophages.

3.2 Atherosclerotic plaques in mouse and human show distinct distributions of macrophage subsets

After manually annotating the myeloid population, the macrophage population was taken for further analysis, and annotation as described in Section 2.4.

For the manual annotation of mouse macrophage subtypes, a low resolution of 0.5, and a high resolution of 1.5 were used (Figure S3). DGE analysis, dot plots of marker genes, information from the HPA [26], gene markers from Panglao database [27], and macrophage subtype markers from prior research, were used to annotate clusters. Annotation resulted in the identification of five distinct macrophage subtypes (Figure 3A) with distinct upregulation (Figure 3B), and distri-

	Human	Mouse
Macrophages	70%	67.5%
Monocytes	15.5%	6.6%
DCs	13.8%	24.5%
NK cells	0.8%	-
T-cells	-	1.3%

Table 3: Distribution of myeloid cell types within atherosclerotic human and mouse plaques.

bution of cell types (Figure 3C, Figure 3D). The inflammatory macrophages exhibited DGE of *Junb*, *Cd74*, and *Cxcl2*, which are all pro-inflammatory marker genes [10]. The resident-like macrophages exhibited anti-inflammatory marker genes *Pf4*, *Cbr2*, and *Folr2*, suggesting accurate annotation of the resident-like macrophage population. Moreover, the DGE of iLAMs did not show any subtype-specific marker genes. However, the dot plot (Figure 3E) shows upregulation of iLAM marker genes *Fabp5*, *Il1b*, and *Plin2*, while lacking upregulation of other subtype-specific marker genes, suggesting that it likely is an iLAM population. In contrast, the LAM population exhibited DGE of known marker genes such as *Lgals3*, *Lpl*, and *Spp1*. Lastly, the IFNIC macrophage population exhibited upregulation of IFNIC marker genes *Irf7*, *Ifit3*, *Ifit1*, *Irf7*, and *Isg15* [8], which lacked upregulation in all the other subtypes (Figure 3E), supporting the IFNIC macrophage annotation.

For the manual annotation of human macrophage subtypes, a low resolution of 0.3, and a high resolution of 0.8 was used for clustering (Figure S6). Annotation was performed similarly to that of mouse macrophage subtypes, leading to the identification of six distinct macrophage subtypes (Figure 4A), and distributions (Figure 4C, Figure 4D). While DGE of the identified inflammatory macrophages exhibited the pro-inflammatory transcript *Cx3cr1* [10], the remaining differentially expressed genes were not typical marker genes. However, prior research has associated *FCGBP*, *IGSF21*, and the remaining differentially expressed genes with inflammatory macrophages. Additionally, the dot plot showed upregulation in pro-inflammatory transcripts *IL1B* and *Cxcl2*, thereby supporting the inflammatory macrophages annotation. The annotated resident-like macrophage population exhibited DGE of *SELENOP*, *F13A1*, and *CD163L1*, which are known resident-like macrophage marker genes [10]. Similarly, the LAM population exhibited known marker genes, such as *APOC1*, *APOE*, and *APOC2*. The iLAM population expressed iLAM marker genes *VCAN*, and *CCL20* [9]. Furthermore, the heatmap and dot plot (Figure 4B, Figure 4E) revealed upregulation of IFNIC marker genes, thereby supporting the IFNIC macrophage annotation. Lastly, a proliferating macrophage population was identified by upregulation of proliferating marker genes (Figure 4E), and DGE of *BIRC5*, and *MKI67* [35] (Figure 4B).

Annotation revealed an absence of proliferating macrophages in mouse samples. In contrast, the proliferating macrophages were consistently present in human carotid samples (Figure 4C), suggesting a possible species-specific difference that may contribute to differing macrophage proliferation mechanisms. However, this absence could be caused due to a lack of macrophage cells in mouse data. According to a tool from the Satija lab[36], if the mouse data are assumed to include six cell types, the rarest cell type population is present for at least 1%, and contains at least 5 cells, then there is a probability of at least 99% that the data contains five proliferating macrophages. However, if this were the case, then these macrophages would presumably be masked by the expression of another subtype in the same cluster due to the low amount of cells.

3.3 Evaluation of macrophage subtype distributions across species

Evaluation of the proportions of macrophage subtypes across species reveals that 20-week-old mice on an atherogenic diet (Atherogenic diet, had the highest number of inflammatory macrophages, followed by mice on a chow diet, and lastly 10-11-week-old mice on an atherogenic diet. Younger mice on an atherogenic diet had the least amount of resident-like macrophages, and the highest amount of LAMs, iLAMs and IFNIC macrophages. It should also be noted that plaques of 10-11-week-old mice on an atherogenic diet consisted of 16.2% iLAMs, whereas the plaques of the 20-week-old-mice on an atherogenic diet consisted of 3.7%, and mice on a chow diet of 0.6% iLAMs. However, the mice on a chow diet consisted of the most resident-like macrophages compared to all other diets (Table 4, Figure 4C).

In total, mouse plaques consisted of 43.2% inflammatory macrophages, 25.5% resident-like macrophages, 21.4% LAMs, 7.8% iLAMs and 1% IFNIC macrophages. In comparison, the human plaques contained a higher proportion of inflammatory macrophages (+5%), iLAMs (+13.2%), IFNIC macrophages (+3.8%) and proliferating macrophages (+1.4%), while having fewer resident-like macrophages (-6.3%) and LAMs (-7.2%).

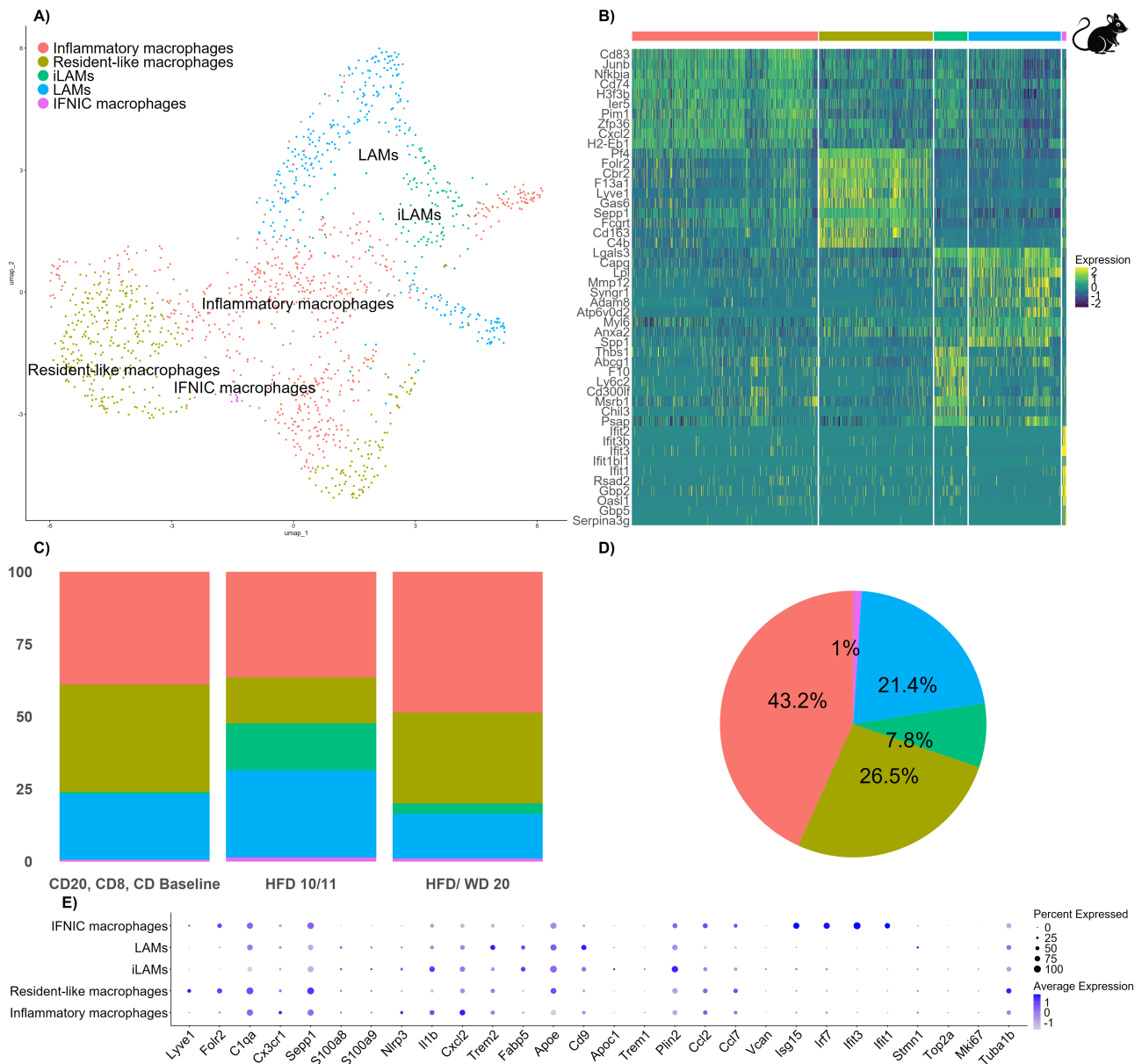


Figure 3: Results of clustering and annotating subtypes of the mouse macrophage population. (A) UMAP visualization of the final annotation of each cluster. (B) Heatmap of major differentially expressed genes of each macrophage subtype. (C) Barplot visualization of the cell type proportions (in percentages) for each condition. (D) Piechart of the distribution of macrophage subtypes in total. (E) Dotplot of known marker genes for macrophage subtypes. These genes are based on current literature [8–10].

Among all conditions, older mice on an atherogenic diet resembled similar levels of inflammatory macrophages compared to human plaques, while younger mice on an atherogenic diet showed the most resemblance in the proportions for resident-like macrophages, iLAMs and IFNIC macrophages. In contrast, mice on a chow diet were least similar to human plaques in terms of distribution.

3.4 Comparative analysis of differentially expressed genes

This section evaluates the results of DGE analysis, identifying genes that are strongly upregulated in one subtype compared to all the other subtypes. Differences in DGE analysis are compared across species, and species-specific genes are subsequently identified.

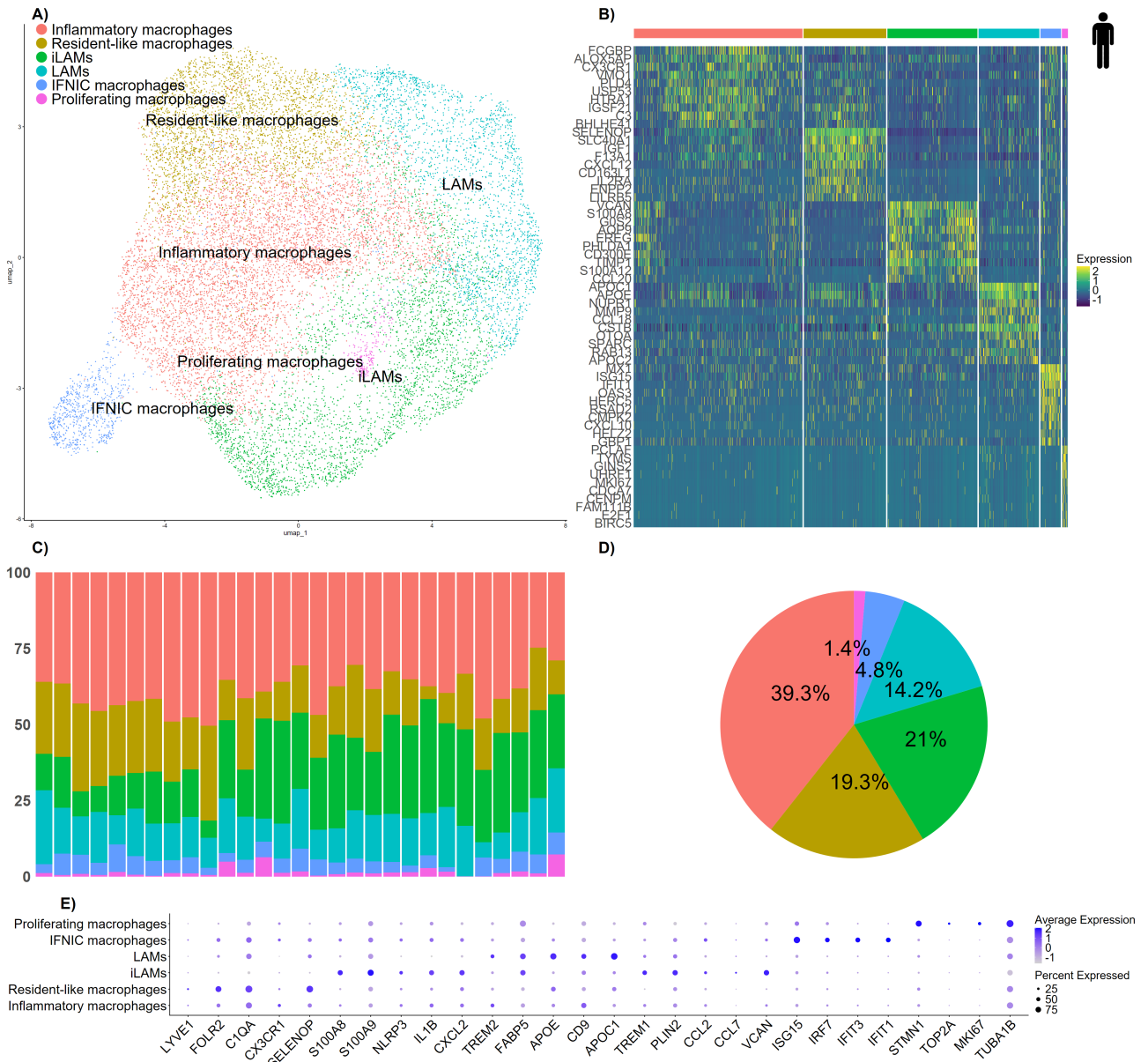


Figure 4: Results of clustering and annotating subtypes of the human macrophage population. (A) UMAP visualizations of the final annotation of each cluster. (B) Heatmap of major differentially expressed genes of each macrophage subtype. (C) Barplot visualization of the cell type proportions (in percentages) for each sample. (D) Piechart of the distribution of macrophage subtypes in total. (E) Dotplot of known marker genes for macrophage subtypes. These genes are based on current literature [8–10].

3.4.1 Species-specific marker genes of resident-like macrophages in mouse and human

Comparison of DGE analysis between the resident-like macrophage population across species revealed few genes that were strongly upregulated in mouse compared to human, particularly *CD163*, *LTC4S*, and *S100A10* (Figure 5A). *LTC4S* is a gene involved in the leukotriene (LT) pathway and initializes the production of cysteinyl-leukotrienes (CysLT) by encoding leukotrien C4 synthase enzyme. Prior research has shown *LTC4S* and CysLT proteins to be pro-inflammatory, associated with disease progression, and foam cell accumulation [37, 38], suggesting that *LTC4S* expression in the resident-like macrophages of mouse may contribute to a species-specific inflammatory expression pattern. In contrast, *CD163* has been shown to be anti-inflammatory by scavenging hemoglobin:haptoglobin complexes, releasing anti-inflammatory cytokines and is suggested to reduce pro-inflammatory effects of the tumor necrosis factor-like weak inducer of apoptosis (TWEAK) pathway in mouse [39]. However, recent research has shown that *CD163*^{-/-} ApoE^{-/-} mouse develop smaller plaques compared to normal ApoE^{-/-} mouse [40], highlighting possible heterogeneity. Additionally, human *CD163* expression has

	Mouse			Human plaques
	Chow diet	Atherogenic diet, (young)	Atherogenic diet, (older)	
Inflammatory macrophages	38.9%	36.5%	48.7%	39.4%
Resident-like macrophages	37.1%	15.8%	31.3%	19.3%
LAMs	22.8%	30.1%	15.4%	14.2%
iLAMs	0.6%	16.2%	3.7%	21%
IFNIC macrophages	0.6%	1.4%	0.9%	4.8%
Proliferating macrophages	-	-	-	1.4%

Table 4: Distribution of macrophage subtypes across organisms and conditions. The chow diet mice samples include mouse of the ages 8, 12 and 20. Moreover, the atherogenic diet refers to the HFD and WD diets, and the young mice refer to the 10 and week old mice, whereas the older mice refers to the 20 week old mice.

been associated with increased plaque vulnerability and progression [40, 41], suggesting that the role of *CD163* might differ between species and be context-specific. The gene *S100A10* was also found to be upregulated in mouse compared to human, and is involved in macrophage recruitment in response to inflammation [42]. Upregulation of *S100A10* may contribute to the macrophage-dominant immune cell landscape in mouse. Moreover, several distinct genes were upregulated in human resident-like macrophages compared to mouse, such as *HLA-DMB*, *IGF1* and *SGK1* (Figure 5A). Among these genes, *HLA-DMB* may contribute to inflammation [43] and rupture of atherosclerotic plaques. Furthermore, *IGF1* has been shown to have atheroprotective effects in animal models. However, the exact effects and mechanisms in humans remain to be elucidated [44]. Additionally, *SGK1* has been shown to increase monocytes/macrophages migration and promote inflammation in mouse [45]. These findings highlight potential pro-inflammatory expression patterns in human resident-like macrophages that are not present in mouse models.

Additionally, several genes were differentially expressed but could not be mapped to a human ortholog (Figure 5B). This

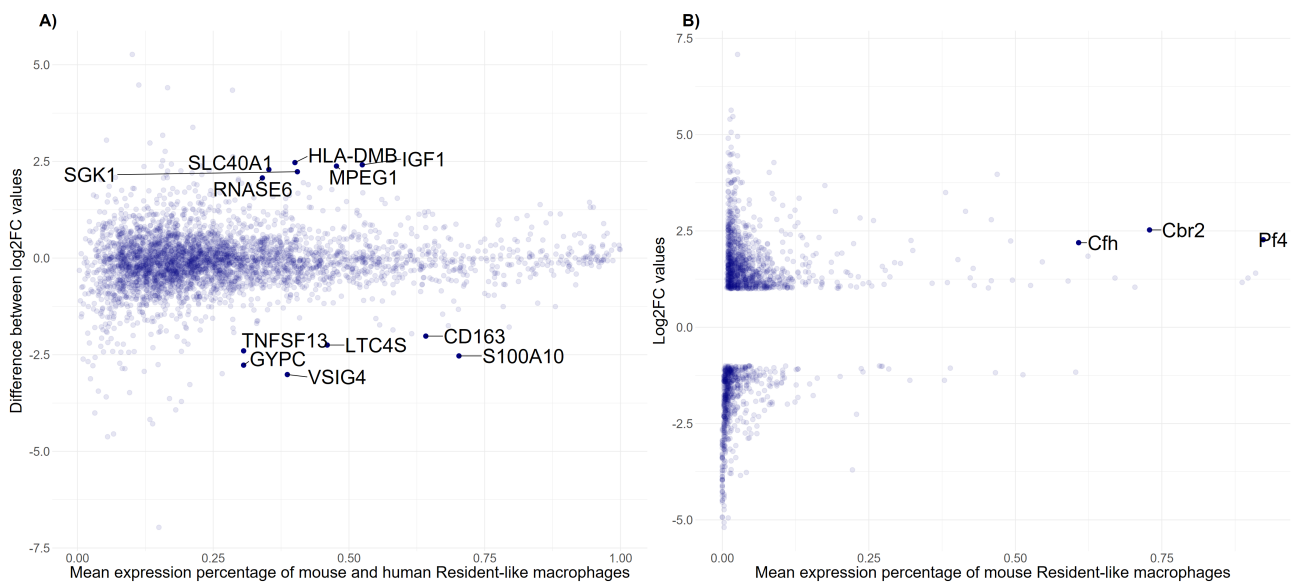


Figure 5: Comparison of differentially expressed genes in resident-like macrophage populations of mouse and human. (A) A plot showing the difference between the log2FC values of mouse and human resident-like macrophage markers on the y-axis, while showing on the x-axis the average expression of marker genes being expressed in the resident-like macrophage population. Genes that have a fold change difference of at least 2, and are on average expressed by 30% of resident-like macrophages across species are visualized. (B) Marker genes in mouse which were not able to be mapped to a human counterpart. On the x-axis is the log2FC value and on the y-axis the average percentage of these genes being expressed in the resident-like macrophage population. Genes that have a fold change of at least 2, and are expressed by 50% of resident-like macrophages in mouse are visualized.

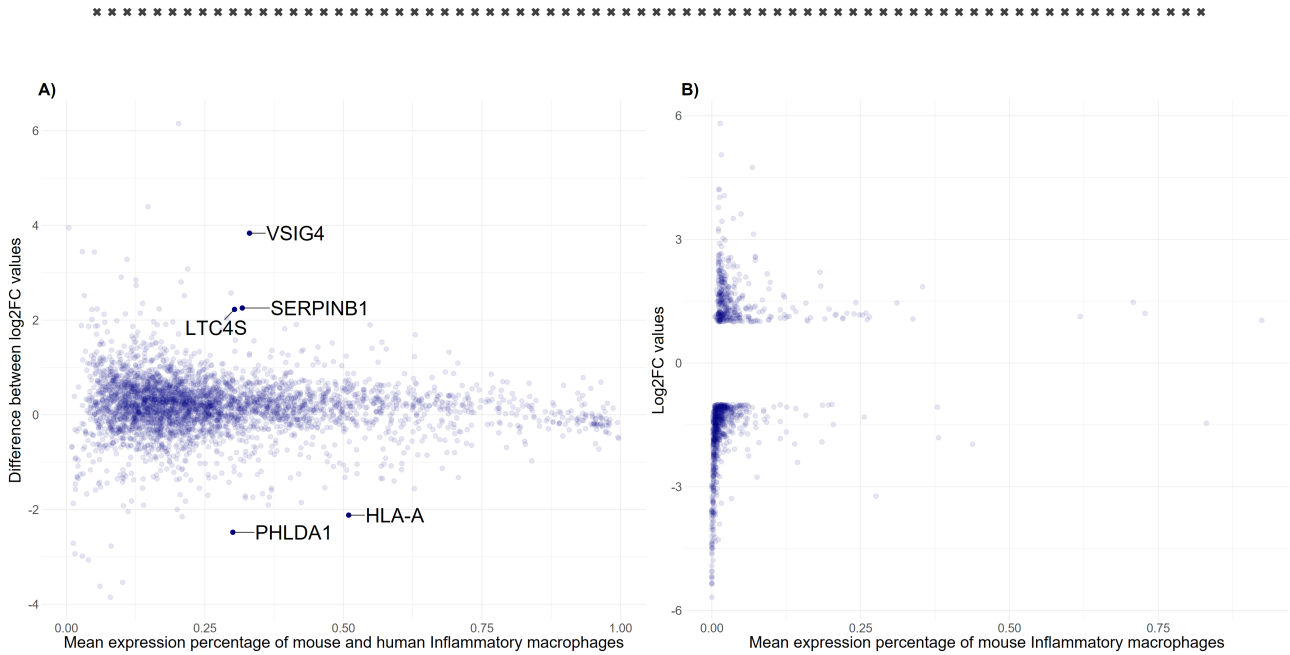


Figure 6: Comparison of differentially expressed genes in inflammatory macrophage populations of mouse and human. (A) A plot showing the difference between the log2FC values of mouse and human inflammatory macrophage markers on the y-axis, while showing on the x-axis the average expression of marker genes being expressed in the inflammatory macrophage population. Genes that have a fold change difference of at least 2, and are on average expressed by 30% of inflammatory macrophages across species are visualized. (B) Marker genes in mouse which were not able to be mapped to a human counterpart. On the x-axis is the log2FC value and on the y-axis the average percentage of these genes being expressed in the inflammatory macrophage population. Genes that have a fold change of at least 2, and are expressed by 50% of inflammatory macrophages in mouse are visualized.

could be caused by a lack of human homolog or extremely low expression in human. Several unmappable genes were *Cfh*, *Cbr2*, and *Pf4*. Among these genes, the human counterparts of *Cfh*, and *Pf4*, were barely expressed in human, whereas no homolog was found for *Cbr2*. The human homolog of *Cfh*, *CFH*, was expressed in only 188 genes, showing a log normalized expression of 0.01, and the human homolog of *Pf4*, *PF4*, was expressed by only 13 cells, showing a log normalized expression of 0.0008. Deletion of *Cfh* or *Pf4* has been associated with decreased disease progression in both mouse as well as human [46–48]. In addition, *Pf4* has been shown to be a marker of resident-like macrophages in mouse [12]. However, the presence of these genes in mouse resident-like macrophages and not human may suggest a species-specific difference. In summary, these differences could contribute to more pro-inflammatory characteristics in mouse resident-like macrophage populations compared to human.

3.4.2 Inflammatory macrophages are largely conserved across species

The inflammatory macrophage subtypes in both mouse and human shared similar marker genes, suggesting conservation (Figure 6A). Additionally, all strongly upregulated genes in mouse were mapped to their human counterpart (Figure 6B). However, *HLA-A*, *LTC4S*, *PHLDA1*, *SERPINB1*, and *VSIG4* showed distinct expression across species. Notably, *PHLDA1* was upregulated in mouse (Log2FC = 1.5, expressed in 45% of inflammatory macrophages), whereas *PHLDA1* was downregulated in human (Log2FC = -1, expressed in 15.2% of inflammatory macrophages). This gene is part of the *PHLDA* family, and has been shown to suppress pro-inflammatory cytokines [49], inhibit the Akt pathway [50], which is a pathway involved in plaque formation and disease progression [51, 52], and plays a role in the p53-mediated apoptosis pathway [53]. P53 is associated with decreased atherosclerotic lesions [54], suggesting that *PHLDA1* expression in inflammatory macrophages is atheroprotective. Downregulation of *PHLDA1* in human inflammatory macrophages may contribute to an atherogenic milieu. In contrast, the genes *LTC4S* and *VSIG4* were downregulated in mouse (Log2FC = -1.3, expressed in 15% of inflammatory macrophages, and Log2FC = -3.3, expressed in 1% of inflammatory macrophages, respectively), and upregulated in human (Log2FC = 1, expressed in 45.7% of inflammatory macrophages, and Log2FC = 0.5, expressed in 65.3% of inflammatory macrophages, respectively). As previously mentioned, prior research has shown that expression of *LTC4S* is associated with pro-inflammatory proteins and disease progression [37, 38], while *VSIG4* is involved in inhibition of pro-inflammatory macrophages [55].

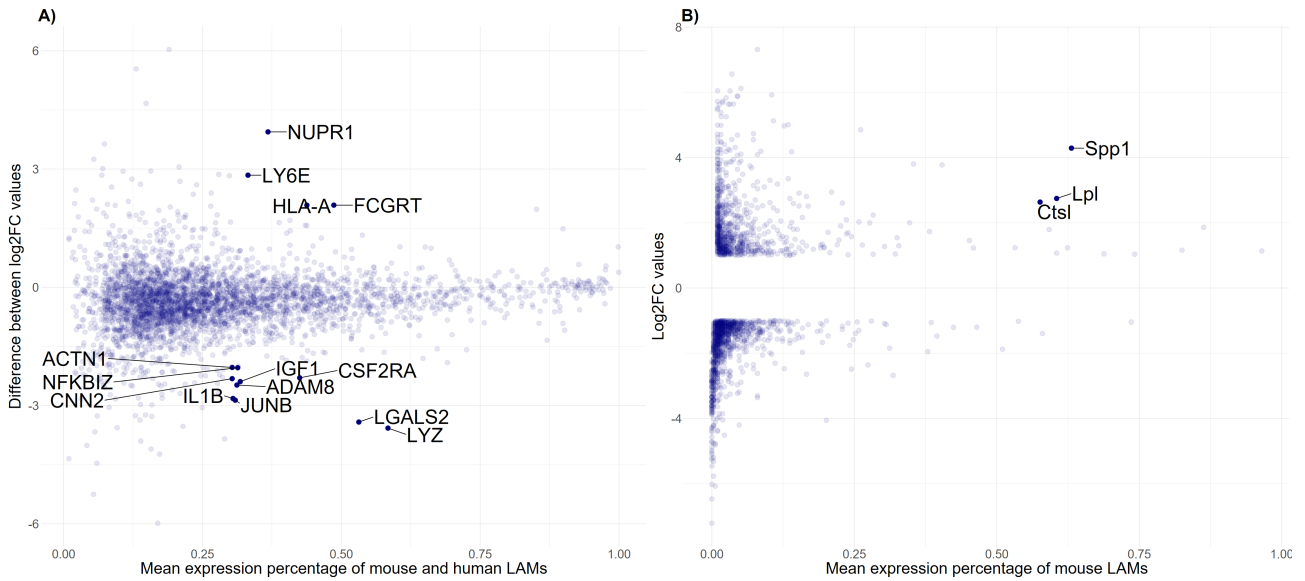


Figure 7: Comparison of differentially expressed genes in LAM populations of mouse and human. (A) A plot showing the difference between the log₂FC values of mouse and human LAM markers on the y-axis, while showing on the x-axis the average expression of marker genes being expressed in the LAM population. Genes that have a fold change difference of at least 2, and are on average expressed by 30% of LAMs across species are visualized. (B) Marker genes in mouse which were not able to be mapped to a human counterpart. On the x-axis is the log₂FC value and on the y-axis the average percentage of these genes being expressed in the LAM population. Genes that have a fold change of at least 2, and are expressed by 50% of LAMs in mouse are visualized.

3.4.3 Comparison of differentially expressed genes between mouse and human LAMs

Comparing DGE analysis across species revealed notable marker genes such as *CSF2RA*, *FCGR2*, *HLA-A*, *LGALS2*, and *LYZ*. Notably *LYZ*, which is the human homolog of *Lyz1*, is upregulated in mouse with a fold change of 2.7, whereas *LYZ* is downregulated in human with a fold change of -0.9. Moreover, Xu et al. (2024) [56] reported *Lyz1* to be exclusively expressed in the LAMs of livers, and exhibit distinguishable expression of the genes *Apoc1*, *Ear2*, and *Id3*. However, these claims contradict our current research, because *Lyz1* is expressed in the LAMs aortic mouse samples, and the LAMs in this study do not show distinguishably gene expression of those genes. Among the genes *Apoc1*, *Ear2*, and *Id3*, *Apoc1* was not statistically significant differentially expressed in LAMs, highlighting that *Apoc1* is not a distinguishable gene in mouse LAMs.

Additionally, the genes *Ctsl*, *Lpl*, and *Spp1* were not mapped to a human homolog, as *LPL* and *CTSL* were barely expressed in human LAMs, and *Spp1* was incorrectly mapped. *LPL* belongs to the lipase gene family, and plays a crucial role in lipoprotein metabolism by facilitating lipid uptake and accumulation. Increased expression of *LPL* has been shown to be atherogenic and increase lesion size significantly in mouse [57, 58]. One key activator of *LPL* is the protein *apoC2* [59]. The gene *APOC2* is a marker gene for human LAMs, however, *APOC2* is only expressed in 0.3% of mouse LAMs. Since *APOC2* is a key activator of *LPL* [59] and is mostly absent in mouse, this may suggest a regulatory difference of *LPL* and lipid metabolism between species in LAMs. Additionally, this might suggest that *LPL* expression in human LAMs is more *APOC2* driven compared to mouse LAMs.

3.4.4 Absence of *TREM1* expression in mouse iLAMs compared to human iLAMs

Among known iLAM marker genes, *TREM1* and *PLIN2*, *TREM1* was only expressed by 38 cells in mouse iLAMs with a normalized expression of 0.03. Additionally, the dot plot (Figure 3E) also reveals absence of *TREM1*. Because *TREM1* is known to be involved in the production of pro-inflammatory cytokines and lipid accumulation [60], suggesting that lack of *TREM1* expression in mice might contribute to less pro-inflammatory activity in mouse iLAMs.

Furthermore, several select genes were upregulated in human iLAMs compared to mouse iLAMs (Figure 8A), such as *FLNA*, *LGPMN*, and *LYZ*. *FLNA* was upregulated in human iLAMs with a fold change of 1.7, whereas it was downregulated in mouse iLAMs with a fold change of -0.5. The gene is associated with increased foam cell formation, and

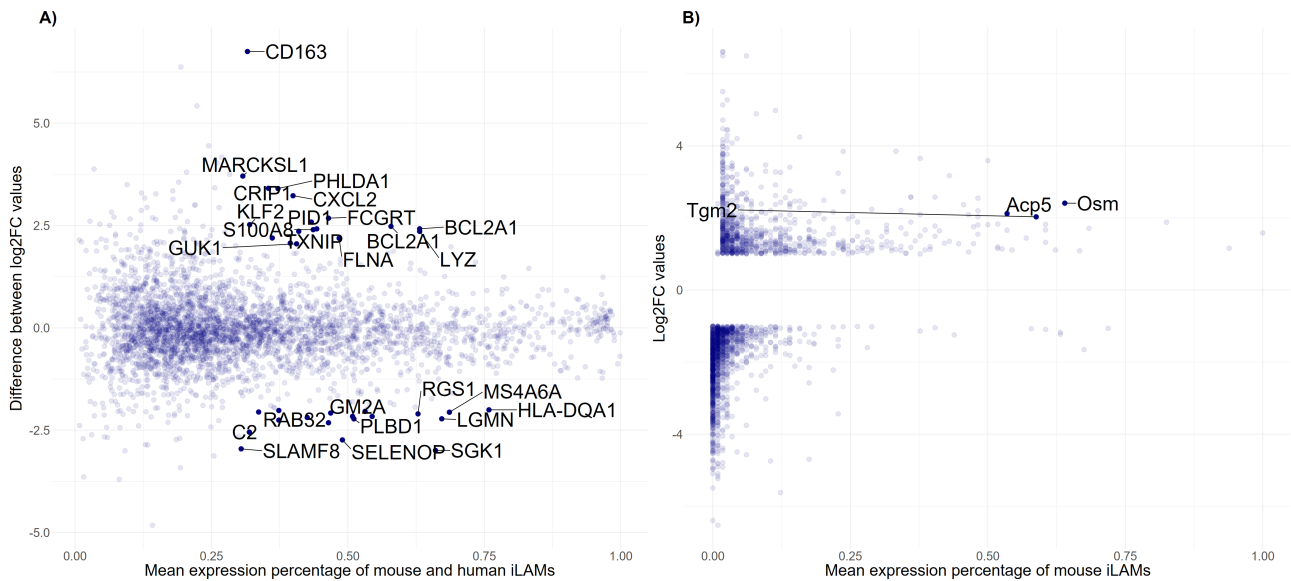


Figure 8: Comparison of differentially expressed genes in iLAM populations of mouse and human. (A) A plot showing the difference between the log2FC values of mouse and human iLAM markers on the y-axis, while showing on the x-axis the average expression of marker genes being expressed in the iLAM population. Genes that have a fold change difference of at least 2, and are on average expressed by 30% of iLAMs across species are visualized. (B) Marker genes in mouse which were not able to be mapped to a human counterpart. On the x-axis is the log2FC value and on the y-axis the average percentage of these genes being expressed in the iLAM population. Genes that have a fold change of at least 2, and are expressed by 50% of iLAMs in mouse are visualized.

promotes atherosclerotic plaque size [61]. Additionally, *LGMN*, which is a gene directly associated with cardiac repair and clearance of inflammatory transcripts [62], was downregulated in human iLAMs and upregulated in mouse iLAMs.

Moreover, the genes *Acp5*, *Osm*, and *Tgm2*, were not mappable to a human ortholog (Figure 8B). *Acp5* and *Osm* are associated with promoting atherosclerotic progression [63, 64], whereas *Tgm2* is associated with an anti-inflammatory phenotype [65].

In summary, human iLAMs upregulated several pro-inflammatory genes, such as *FLNA*, *LGMN*, and *TREM1*, whereas mouse iLAMs downregulated *FLNA*, *LGMN*, and showed low expression of *TREM1*. Additionally, the unmappable genes *Acp5* and *Osm*, were also associated with atherosclerotic disease progression. These findings suggest that human and mouse iLAMs may exhibit a species-specific inflammation pathway. However, to fully support this claim a pathway enrichment should be performed.

3.4.5 IFN γ macrophage populations exhibit mouse-specific genes

Figure 9A shows an abundance of differentially expressed genes that are upregulated or downregulated in mouse compared to human. Notably, the mouse genes *BC094916*, *Ifi205*, *Pydc3*, *Pydc4*, *Pyhin1*, *Mnda*, *Ifi203*, *Ifi204*, *Mnda*, and *AI607873* were mapped to the human gene *IFI16*, suggesting evolutionary divergence which may contribute to species-specific expression patterns. Other upregulated marker genes in mouse were *IFIT3*, *GBP2*, *IFIT2*, *RSAD2*, *LYZ*, *STAT2*, *TOR3A*, and *IFI44*. Among these genes, *IFIT2* was most upregulated in mouse with a fold change of 8.5 compared to human IFN γ ICs (Log2FC = 2.8). Additionally, the mouse genes *Ifit3b*, *Ifit1*, *Ifi47*, and *Oasl2* were not mappable to a human ortholog (Figure 9B). Lastly, transcriptional analyses revealed that mouse IFN γ ICs strongly expressed the transcription factor *Ir7f*, while their human counterpart did not (Figure S7a, S7b). Additionally, *IRF7* is an interferon regulatory factor that regulates type 1 interferon [66]. Expression of type 1 interferon contributes to inflammation signaling [67], and it has been suggested that *IRF7* and *IRF3* interact with each other to restore capillaries and formation of arteries after ischemic stroke [68]. Lack of *IRF7* in human relative to mouse might suggest that mouse exhibit more tissue repair after major cardiovascular events. However, this suggestion lacks statistical power as the mouse IFN γ IC population contains only 15 cells.

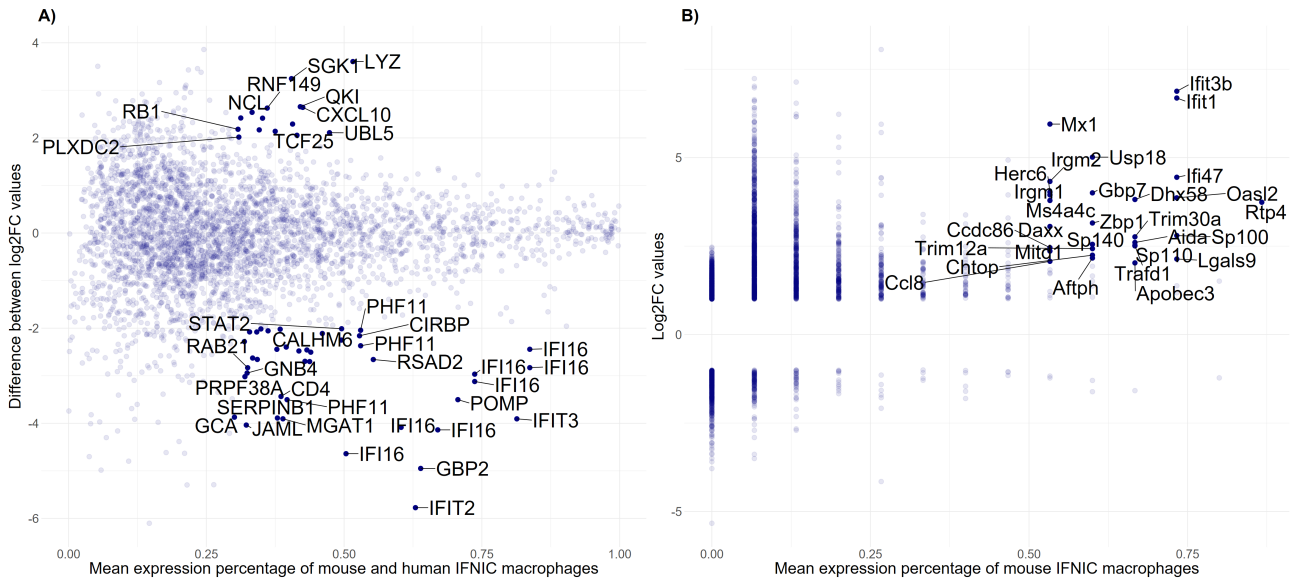


Figure 9: Comparison of differentially expressed genes in IFNIC populations of mouse and human. (A) A plot showing the difference between the log2FC values of mouse and human IFNIC macrophage markers on the y-axis, while showing on the x-axis the average expression of marker genes being expressed in the IFNIC macrophage population. Genes that have a fold change difference of at least 2, and are on average expressed by 30% of IFNIC macrophages across species are visualized. (B) Marker genes in mouse which were not able to be mapped to a human counterpart. On the x-axis is the log2FC value and on the y-axis the average percentage of these genes being expressed in the IFNIC macrophage population. Genes that have a fold change of at least 2, and are expressed by 50% of IFNIC macrophages in mouse are visualized.

4 Discussion

The aim of this study was to analyze and compare transcriptional differences in macrophage and macrophage subtypes between atherosclerotic mouse and human macrophages to evaluate the translational value of using mouse models in atherosclerosis research. To achieve this, mouse and human scRNA-seq data was analyzed using Seurat V5, and the results were compared. These results suggest that 10-11-week-old mice on an atherogenic diet show comparable proportions of macrophage subtypes compared to human. However, the human plaques show more LAMs and IFNIC macrophages, suggesting that human plaques are more inflammatory compared to the 10-11-week-old mouse plaques. The mouse resident-like macrophage population exhibited upregulation of *S100A10*, which may contribute to increased macrophage recruitment. Furthermore, *Cfh*, and *Pf4*, were only differentially expressed in mouse resident-like macrophages. Upregulation of *Cfh*, *Pf4*, and *S100A10* in mouse resident-like macrophages may contribute to more pro-inflammatory characteristics compared to human resident-like macrophage populations. Moreover, the inflammatory macrophage populations between species are largely conserved. In contrast, iLAMs showed a difference in expression of pro-inflammatory genes *FLNA*, *LGMN*, and *TREM1*, possibly suggesting an increased inflammatory characteristic in human iLAMs compared to their mouse counterpart. Furthermore, mouse LAMs showed increased expression of *Lpl* and decreased expression of its co-factor *apoC2*, which may suggest differences in lipid handling between mouse and human LAM populations. Lastly, the IFNIC macrophage population exhibited an evolutionary divergence and possible species-specific expression patterns. In conclusion, most macrophage subtypes, besides IFNIC macrophages, shared transcriptional similarities across species. Only select genes showed distinct expression patterns.

However, this study has several methodological limitations. First of all, the integration on the mouse plaques has been performed on condition, rather than per sample. Because of this, batch effects in conditions containing samples from distinct studies could result in suboptimal removal of batch effects in the mouse data. Secondly, no benchmarking method, such as scIB [69], has been performed on the integration methods, giving no indication on the quality of integration. Thirdly, there is a significant imbalance in dataset size between the mouse and human data. It has been suggested that some integration methods showed increased quality of integration with reduced dataset imbalances [70]. Among the benchmarked integration methods, Seurat V4 CCA integration exhibited an increase in integration quality when more data was available, regardless of dataset imbalances. However, currently there is no study performed studying the effect of dataset imbalances using Seurat V5 CCA integration. Since the results of a Seurat V4 CCA integration and a Seurat V5 CCA integration are not identical [71], the exact effect of this imbalance is unknown, although it is likely that the effect

on dataset imbalances will be similar. Another limitation is that no cross-species integration method has been applied, leading to less comparable analyses, which decreases the validity of the results. Lastly, inflammatory macrophages were shown to be largely conserved. However, both mouse as well as human inflammatory macrophages populations showed relatively low upregulation of known marker genes. One possible explanation is that some inflammatory macrophages were annotated as iLAMs instead, leading to lower expression of typical inflammatory macrophage marker genes.

Despite the limitations in the current study, our findings align with those of Zernecke et al., which reported transcriptional similarity in macrophages across species [12]. Especially the inflammatory and resident-like macrophage populations shared large conservation across species. Overall, these findings support the translatability of mouse models to human atherosclerosis research, especially in macrophage function.

For future research, it will be important to gain access to larger scRNA-seq datasets of atherosclerotic mice. Due to a lack of macrophage cells ($n = 1,464$) in the mouse immune cell population, the samples had to be merged based on condition instead of per sample. As described above, this could lead to a suboptimal removal of batch effects. The study performed by Zernecke et al. [12] reported a similar limitation, also highlighting the low numbers of macrophages in some of their samples. To address this limitation, future research should increase the amount of plaques gathered per batch when pooling. Additionally, future cross-species analysis should evaluate the integration quality between samples by applying a benchmarking method such as scIB [69]. This would lead to increased validity of the results, and would give insights into possible artifacts caused by low quality integration. Another method to improve integration quality, is to perform a cross-species integration method that does not rely on orthology mapping. A non-orthologous deep learning mapping approach, such as SATURN or SAMap [72, 73], would group functionally related genes, even if there is no known homology. This would highlight species-specific differences, leading to insights that could refine existing models by knockout of species-specific genes, resulting in improved translatability of mouse models in human atherosclerosis research.

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6 Supplementary material

6.1 Gene expression omnibus of available datasets

* * * * *

Study	Organism	GSE
Cochaine et al. (2018)	Mouse	GSE97310
Vafadarnejad et al. (2020)	Mouse	GSE97310
Winkels et al. (2018)	Mouse	GSE245373, GSE149069
Horstmann et al. (2024)	Human	GSE245373
Fernandez et al. (2019)	Human	GSE22473
Dib et al. (2023)	Human	GSE210152
Bashore et al. (2024)	Human	GSE253903

Table S1: Overview of gene expression omnibus accession codes of the publicly available scRNA-seq datasets used.

* * * * *



6.2 Cluster plots

This subsection shows the cluster plots on several levels (All immune cells, myeloid subset, and macrophage subset) for different resolutions and both species.

6.2.1 Cluster plots of all mouse immune cells

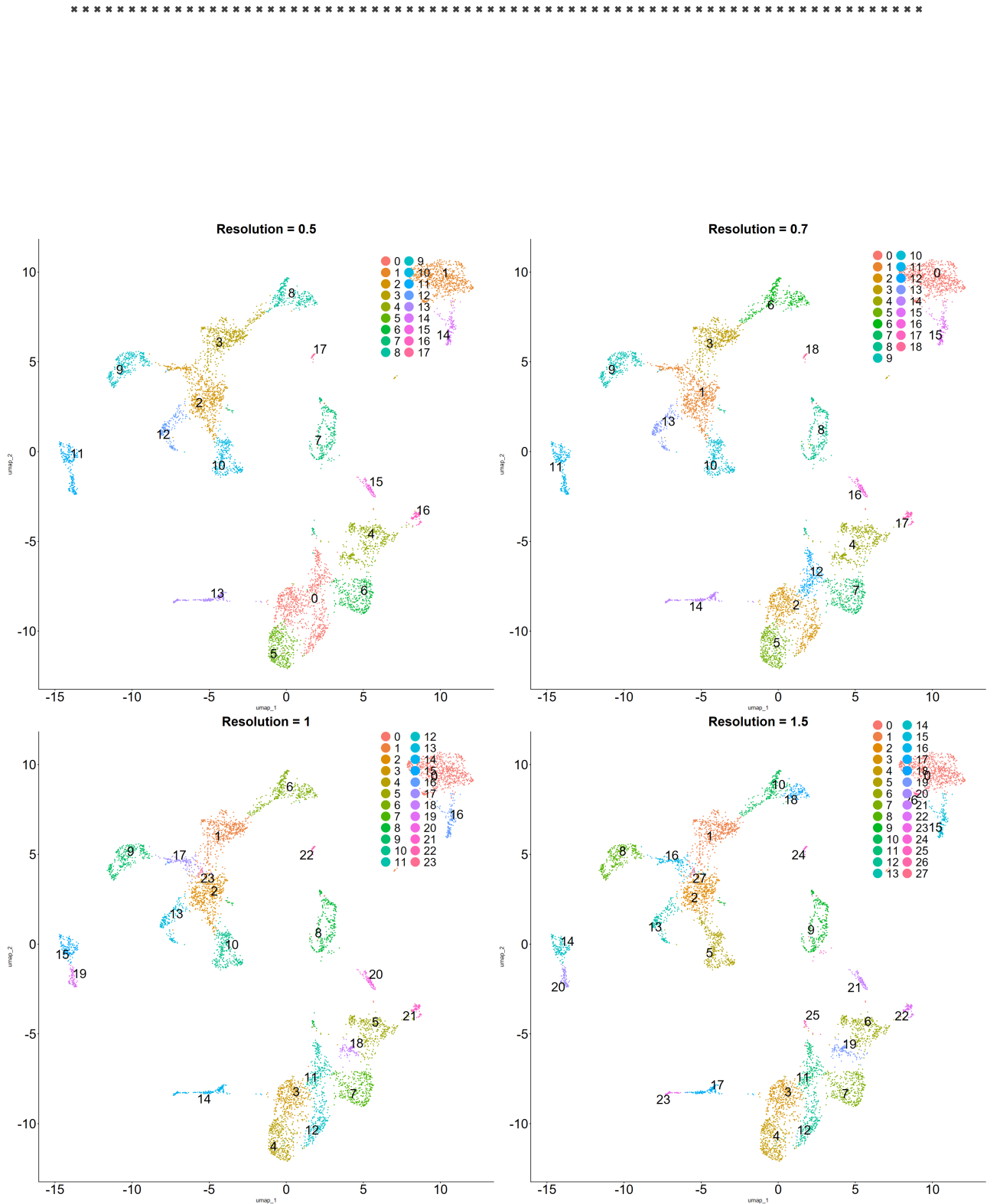


Figure S1: Result of clustering all mouse immune cells using several resolutions and the “FindClusters” function, visualized in a UMAP.

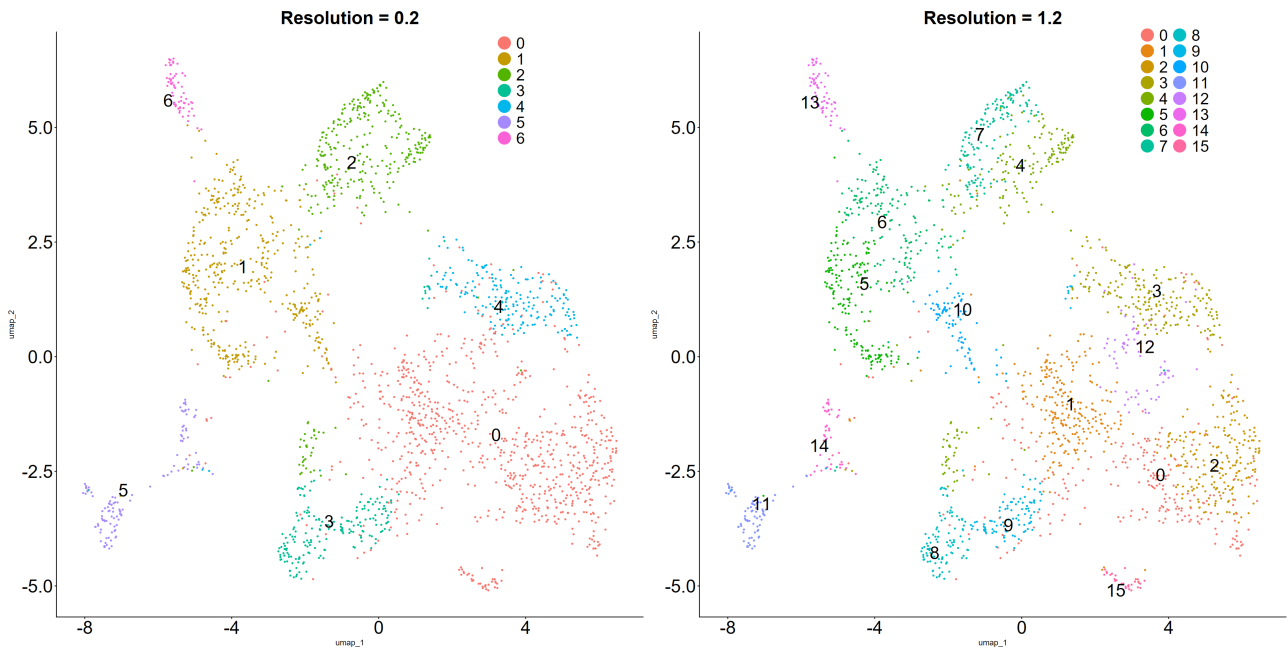


Figure S2: Result of clustering mouse myeloid cell population using several resolutions and the “FindClusters” function, visualized in a UMAP plot.

6.2.2 Cluster plots of mouse myeloid subset

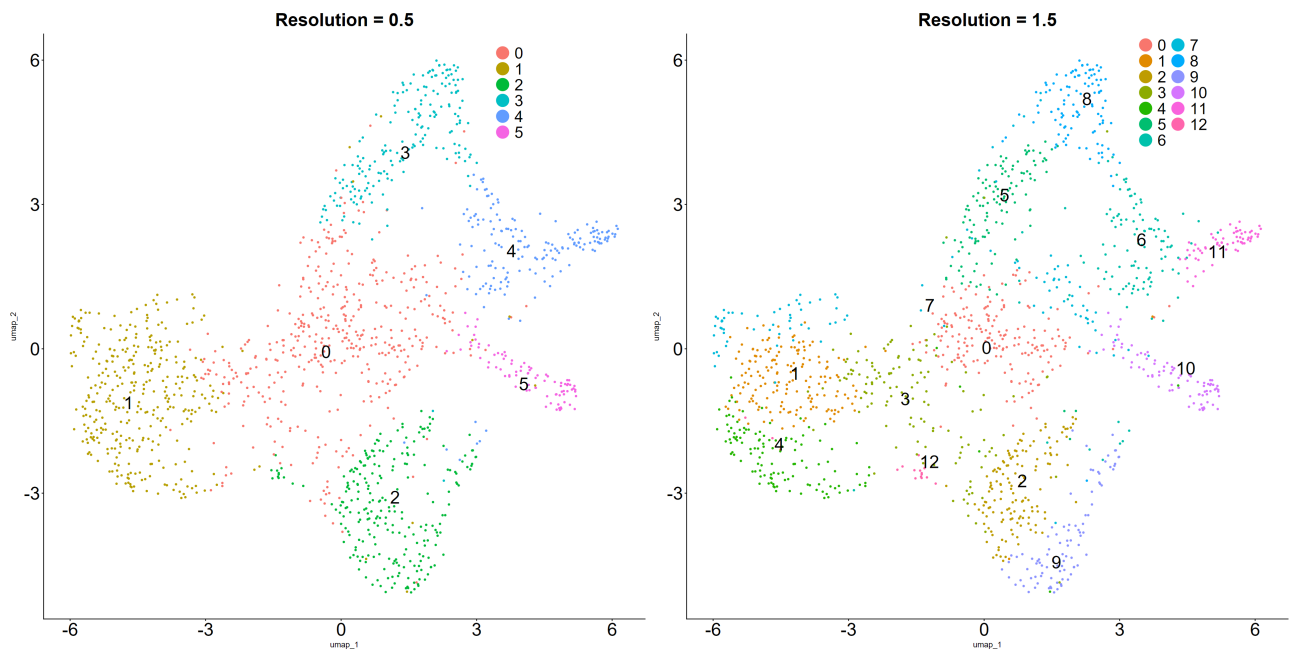


Figure S3: Result of clustering mouse macrophage cell population using several resolutions and the “FindClusters” function, visualized in a UMAP plot.

6.2.3 Cluster plots of mouse macrophage subset



6.2.4 Cluster plots of all human immune cells

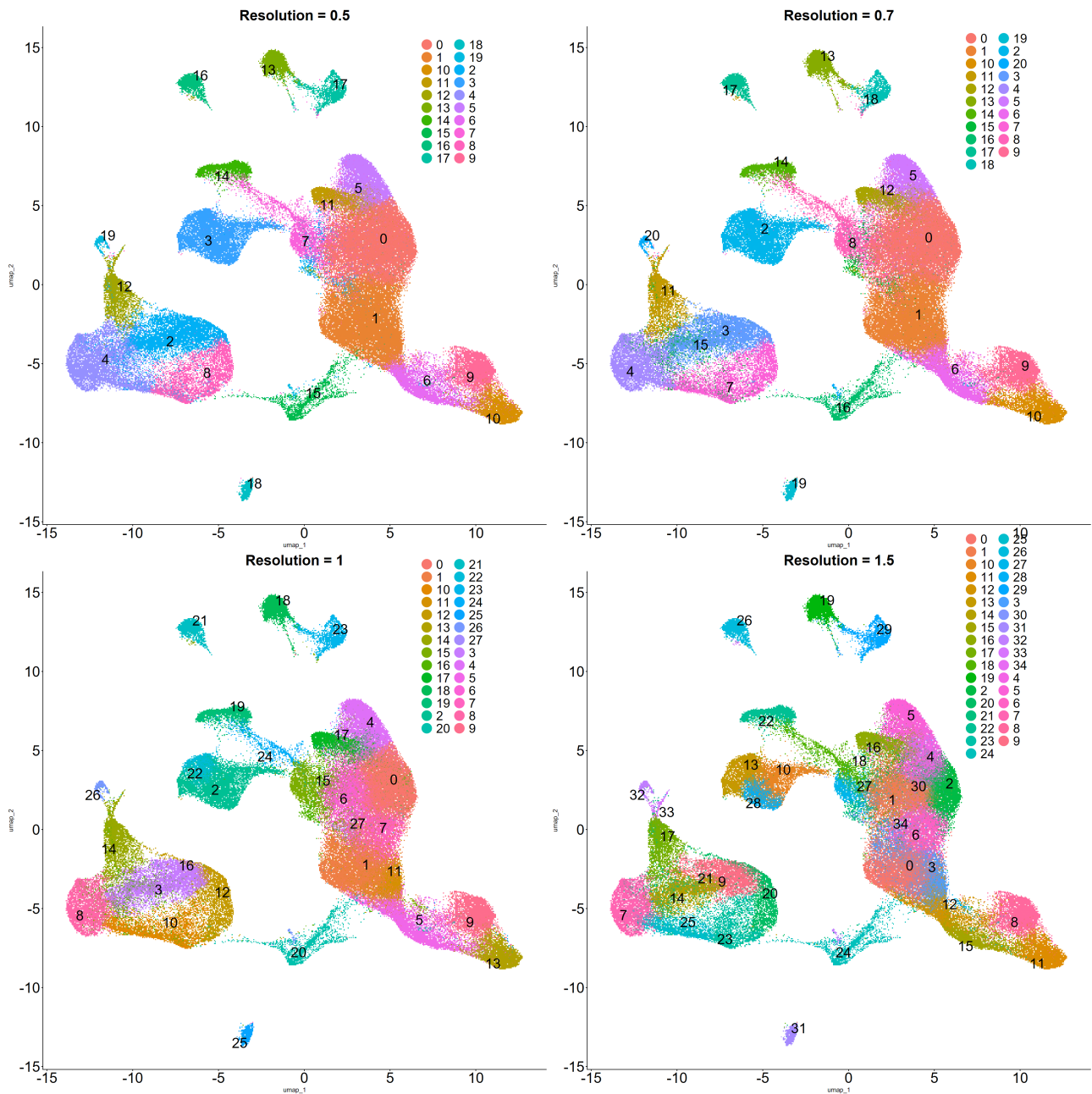
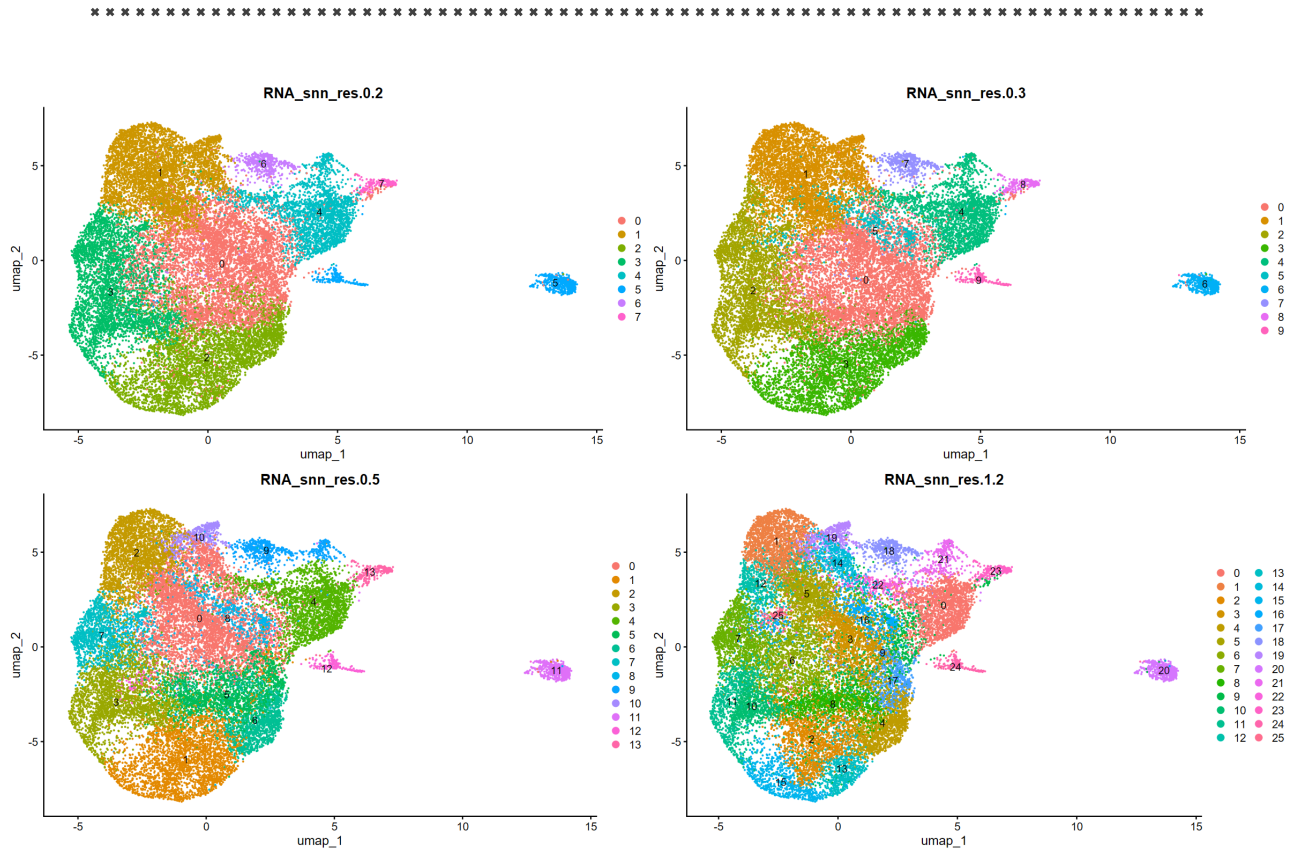
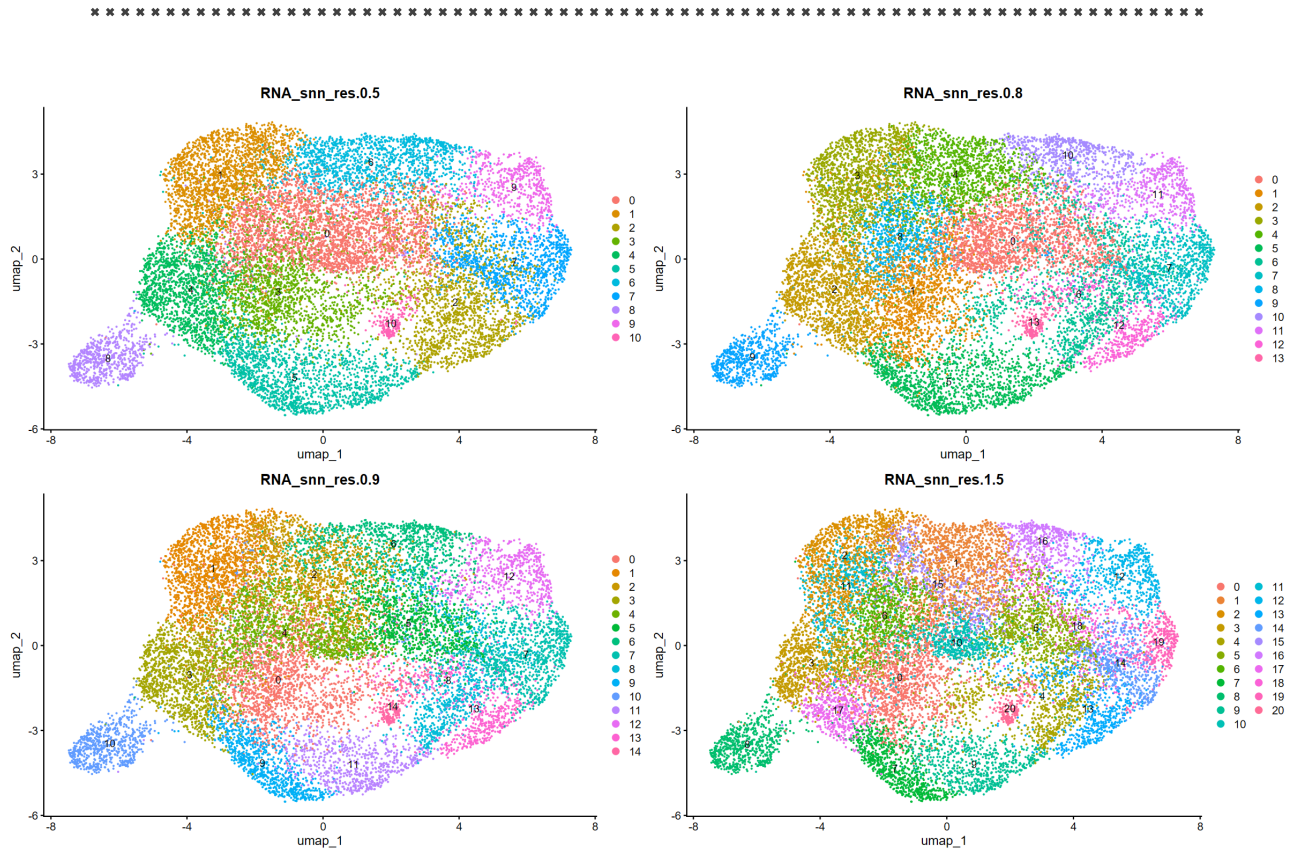


Figure S4: Result of clustering all mouse immune cells using several resolutions and the “FindClusters” function, visualized in a UMAP plot.



6.2.5 Cluster plots of human myeloid subset



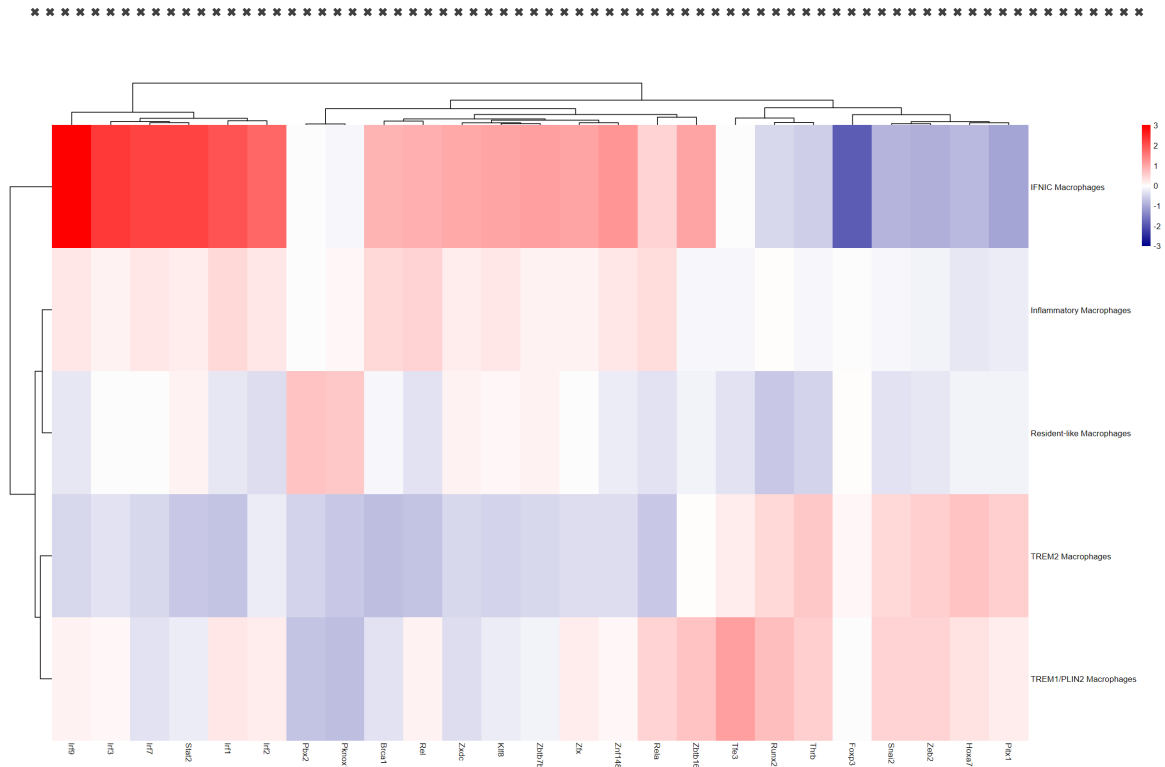
6.2.6 Cluster plots of human macrophage subset

6.3 Annotation of macrophage subtypes

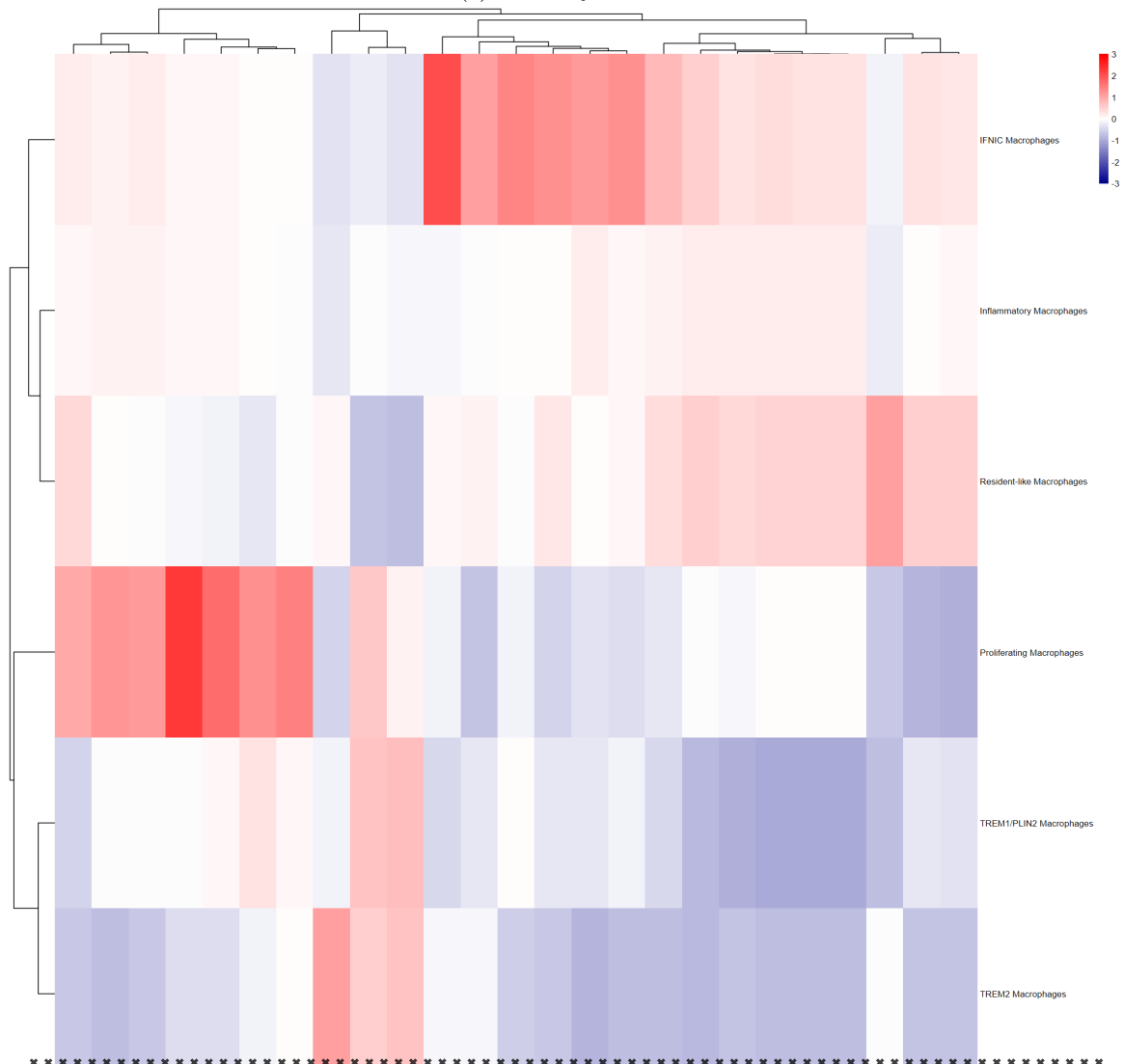
High resolution (resolution = 1.5) clustering of mouse macrophage populations revealed 13 distinct clusters, which are visualized using UMAP (Figure S3). Clusters 0, 2, 3 and 7 showed DGE of inflammatory transcripts, such as *Il1b*, *Ccl4*, *Ccl12*, *Egr1*, *Dnajb1*, *H2-Ab1*, and *Cd74*, suggesting that these clusters belong to an inflammatory macrophage population [10]. In contrast, clusters 1 and 4 differentially express anti-inflammatory transcripts, such as *Ccl8*, *Cd163*, *Lyve1*, *Ccl24*, *Folr*, *Fxyd2*, and *Ltc4s*. Additionally, dot- and violin plots of known resident-like marker genes revealed upregulation in clusters 1, 4 and 9, indicating that these clusters belong to a resident-like macrophage population. Furthermore, clusters 5 and 8 showed upregulation for LAM markers (Figure 3E). Also, clusters 8 exhibited differential expression for the genes *Fabp5*, *Ctsl*, and *SPP1*, which are typical LAM marker genes [10]. Additionally, cluster 9 expressed no distinct gene markers. However, when using the “FindMarkers” against known inflammatory macrophage and resident-like macrophage populations, it was found that LAM marker genes *CD9* and *SPP1* were consistently differentially expressed, indicating that cluster 9 also belongs to a LAM population. Although, cluster 6 also did not exhibit any known marker genes during the differential expression analysis, dot plots revealed upregulation for several iLAM marker genes, suggesting cluster 6 to belong to an iLAM population. Lastly, cluster 12 exhibited differential expression of *Gbp2*, *Ifit3b*, *Ifit1*, and *Ifit2* and increased upregulation for known IFNIC marker genes (Figure 3E), indicating cluster 12 to be an IFNIC population.

High-resolution (resolution = 0.8) clustering of human macrophage populations revealed 14 clusters, which are visualized using a UMAP (Figure 4). Clusters 0, 1, 2, and 8 exhibited differential expression of marker genes *VMO1*, *ALOX5AP*, *FCGBP*, *LYZ*, *JAML*, *FCN1*, *CLEC10A*, *GPR34*, *USP53*, *IGSF21* and *CX3CR1*. Previous annotation of atherosclerotic human macrophage subtypes has shown these marker genes to be associated with inflammatory macrophages. Moreover, the clusters 3 and 4 differentially expressed the genes *SELENOP*, *F14A1*, *CD163L1* and *CD209*, indicating a resident-like macrophage population [10]. Additionally, typical resident-like macrophage marker genes such as *FOLR2* and the *C1Q* family were upregulated as seen in the dotplot (Figure 4). Clusters 5, 6 and 12 showed upregulation of the iLAM marker genes *TREM1*, *PLIN2*, and inflammatory transcripts in the dot plot, indicating an iLAM population. In addition, differential expression analysis revealed expression of *VCAN*, *S100A8*, *S100A12*, and *PLIN2*, which are iLAM-associated genes [9]. Another subset of clusters 7, 10, and 11, exhibited differential expression of the genes *MMP9*, *LPL*, *APOC1*, and *APOE*. Additionally, cluster 7 and 10 showed upregulation for *TREM2*, *CD9*, *FABP5*, *APOE*, and *APOC1*, indicating that these clusters belong to a LAM population. Two clusters were left to identify, cluster 9 and 13. Cluster 9 exhibited differential expression of the genes *ISG15*, *MX1*, *IFIT1*, which are all marker IFNIC-associated marker genes, indicating that this cluster belongs to an IFNIC population [8]. Lastly, cluster 13 exhibited differential expression of the genes *PCLAF*, *TYMS*, *GINS2*, *UHRF1*, *MKI67*, *CDCA7*, and *BIRC5*. The genes *MKI67* and *BIRC5* are associated with proliferating macrophages [8, 35]. In addition, the dot plot shows upregulation of the proliferating macrophages marker genes *STMN1*, *TOP2A*, *MKI67* and *TUBA1B* [8], indicating that cluster 13 belongs to a proliferating macrophages population.

6.4 Transcription factor analysis



(a) TF analysis mice



(b) TF analysis humans

6.5 Raw result of mouse and human differential expression analysis

6.5.1 Mouse macrophage gene markers

cluster	gene	avg_log2FC	pct.1	pct.2	p_val	p_val_adj
Inflammatory Macrophages	Tnfsf9	1.85	0.35	0.16	0.00	0.00
Inflammatory Macrophages	Dusp2	1.82	0.32	0.15	0.00	0.00
Inflammatory Macrophages	Nlrp3	1.68	0.35	0.17	0.00	0.00
Inflammatory Macrophages	Kdm6b	1.63	0.53	0.30	0.00	0.00
Inflammatory Macrophages	Maff	1.63	0.27	0.14	0.00	0.00
Inflammatory Macrophages	Cd83	1.62	0.82	0.63	0.00	0.00
Inflammatory Macrophages	Rab20	1.54	0.39	0.23	0.00	0.00
Inflammatory Macrophages	Ccl3	1.47	0.71	0.50	0.00	0.00
Inflammatory Macrophages	Phlda1	1.46	0.45	0.26	0.00	0.00
Inflammatory Macrophages	Sde2	1.46	0.31	0.19	0.00	0.00
Inflammatory Macrophages	Cxcl2	1.46	0.79	0.64	0.00	0.00
Inflammatory Macrophages	Il1b	1.45	0.66	0.45	0.00	0.00
Inflammatory Macrophages	Ccl4	1.40	0.74	0.60	0.00	0.00
Inflammatory Macrophages	Nr4a1	1.40	0.63	0.38	0.00	0.00
Inflammatory Macrophages	Tnfaip3	1.36	0.47	0.27	0.00	0.00
Inflammatory Macrophages	Rel	1.35	0.45	0.28	0.00	0.00
Inflammatory Macrophages	Marcksl1	1.34	0.63	0.43	0.00	0.00
Inflammatory Macrophages	Ifrd1	1.34	0.63	0.43	0.00	0.00
Inflammatory Macrophages	Bcl2a1a	1.34	0.28	0.18	0.00	0.00
Inflammatory Macrophages	Bcl2a1d	1.33	0.48	0.36	0.00	0.00
Inflammatory Macrophages	Skil	1.31	0.48	0.30	0.00	0.00
Inflammatory Macrophages	Tgif1	1.30	0.50	0.35	0.00	0.00
Inflammatory Macrophages	Ccrl2	1.30	0.59	0.37	0.00	0.00
Inflammatory Macrophages	Dnajb1	1.30	0.39	0.32	0.00	1.00
Inflammatory Macrophages	Tlr2	1.28	0.57	0.35	0.00	0.00
Inflammatory Macrophages	Csrnp1	1.28	0.39	0.22	0.00	0.00
Inflammatory Macrophages	Tnf	1.27	0.41	0.24	0.00	0.00
Inflammatory Macrophages	Traf1	1.21	0.26	0.16	0.00	0.01
Inflammatory Macrophages	Birc3	1.21	0.34	0.22	0.00	0.00
Inflammatory Macrophages	Pim1	1.21	0.73	0.49	0.00	0.00
Resident-like Macrophages	Lyve1	3.97	0.47	0.04	0.00	0.00
Resident-like Macrophages	Ccl24	3.50	0.38	0.06	0.00	0.00
Resident-like Macrophages	Cd163	3.31	0.46	0.08	0.00	0.00
Resident-like Macrophages	Fxyd2	3.01	0.41	0.10	0.00	0.00
Resident-like Macrophages	Ltc4s	2.86	0.47	0.12	0.00	0.00
Resident-like Macrophages	Folr2	2.84	0.70	0.16	0.00	0.00
Resident-like Macrophages	C4b	2.79	0.46	0.09	0.00	0.00
Resident-like Macrophages	Selenbp1	2.77	0.26	0.07	0.00	0.00
Resident-like Macrophages	F13a1	2.63	0.77	0.25	0.00	0.00

cluster	gene	avg_log2FC	pct.1	pct.2	p_val	p_val_adj
Resident-like Macrophages	Cbr2	2.53	0.73	0.18	0.00	0.00
Resident-like Macrophages	Pf4	2.27	0.92	0.48	0.00	0.00
Resident-like Macrophages	Sult1a1	2.25	0.32	0.08	0.00	0.00
Resident-like Macrophages	Ccl8	2.24	0.49	0.25	0.00	0.00
Resident-like Macrophages	Gas6	2.20	0.69	0.22	0.00	0.00
Resident-like Macrophages	Cfh	2.19	0.61	0.20	0.00	0.00
Resident-like Macrophages	Npl	2.16	0.26	0.06	0.00	0.00
Resident-like Macrophages	Maf	2.13	0.68	0.29	0.00	0.00
Resident-like Macrophages	Fcgrt	2.07	0.79	0.39	0.00	0.00
Resident-like Macrophages	Stard8	1.99	0.32	0.10	0.00	0.00
Resident-like Macrophages	Rnase4	1.84	0.62	0.30	0.00	0.00
Resident-like Macrophages	Clec10a	1.82	0.36	0.16	0.00	0.00
Resident-like Macrophages	Gpr34	1.82	0.38	0.13	0.00	0.00
Resident-like Macrophages	Wwp1	1.81	0.39	0.18	0.00	0.00
Resident-like Macrophages	Stab1	1.79	0.62	0.32	0.00	0.00
Resident-like Macrophages	Ap2a2	1.78	0.59	0.28	0.00	0.00
Resident-like Macrophages	Slco2b1	1.77	0.36	0.13	0.00	0.00
Resident-like Macrophages	Tppp3	1.76	0.40	0.24	0.00	0.00
Resident-like Macrophages	Slc9a9	1.72	0.26	0.11	0.00	0.00
Resident-like Macrophages	Igfbp4	1.70	0.55	0.21	0.00	0.00
Resident-like Macrophages	Tmem37	1.70	0.50	0.27	0.00	0.00
LAM	Atp6v0d2	4.85	0.26	0.03	0.00	0.00
LAM	Spp1	4.29	0.63	0.28	0.00	0.00
LAM	Syng1	3.80	0.35	0.06	0.00	0.00
LAM	Mmp12	3.77	0.40	0.08	0.00	0.00
LAM	Anpep	3.08	0.26	0.05	0.00	0.00
LAM	Gpnmb	3.05	0.32	0.09	0.00	0.00
LAM	Adam8	2.90	0.36	0.06	0.00	0.00
LAM	Lpl	2.75	0.60	0.20	0.00	0.00
LAM	Lyz1	2.66	0.41	0.28	0.00	0.00
LAM	Ctsl	2.64	0.58	0.51	0.00	0.02
LAM	Fnl1	2.58	0.40	0.17	0.00	0.00
LAM	Fabp5	2.49	0.52	0.23	0.00	0.00
LAM	Cd9	2.28	0.70	0.38	0.00	0.00
LAM	Ear2	2.24	0.28	0.06	0.00	0.00
LAM	Pld3	2.05	0.57	0.31	0.00	0.00
LAM	AA467197	2.03	0.26	0.07	0.00	0.00
LAM	Lgals3	2.01	0.97	0.70	0.00	0.00
LAM	2010111I01Rik	2.00	0.35	0.12	0.00	0.00
LAM	Retnla	2.00	0.28	0.20	0.00	1.00
LAM	Capg	1.86	0.86	0.45	0.00	0.00
LAM	Ctsd	1.84	0.87	0.75	0.00	0.00
LAM	S100a1	1.79	0.59	0.30	0.00	0.00

cluster	gene	avg_log2FC	pct.1	pct.2	p_val	p_val_adj
LAM	Gpr137b	1.77	0.47	0.23	0.00	0.00
LAM	Vat1	1.73	0.38	0.16	0.00	0.00
LAM	Kcnn4	1.70	0.37	0.13	0.00	0.00
LAM	Anxa1	1.69	0.75	0.41	0.00	0.00
LAM	Dpep2	1.69	0.56	0.23	0.00	0.00
LAM	Plp2	1.68	0.47	0.19	0.00	0.00
LAM	Pfklp	1.64	0.34	0.15	0.00	0.00
LAM	Anxa2	1.61	0.79	0.44	0.00	0.00
iLAM	Chil3	3.85	0.33	0.04	0.00	0.00
iLAM	Ly6c2	3.65	0.38	0.04	0.00	0.00
iLAM	Thbs1	3.59	0.50	0.07	0.00	0.00
iLAM	F10	3.17	0.43	0.06	0.00	0.00
iLAM	Fam20c	3.02	0.35	0.08	0.00	0.00
iLAM	Sgk1	2.94	0.60	0.20	0.00	0.00
iLAM	Ass1	2.58	0.36	0.06	0.00	0.00
iLAM	Fam46c	2.58	0.40	0.10	0.00	0.00
iLAM	Tmem38b	2.58	0.36	0.05	0.00	0.00
iLAM	Bhlhe40	2.57	0.52	0.13	0.00	0.00
iLAM	Gdf15	2.55	0.48	0.16	0.00	0.00
iLAM	Osm	2.41	0.64	0.23	0.00	0.00
iLAM	Cd300lf	2.37	0.46	0.07	0.00	0.00
iLAM	Slc7a11	2.24	0.29	0.06	0.00	0.00
iLAM	Mxd1	2.21	0.37	0.12	0.00	0.00
iLAM	Nr1h3	2.21	0.34	0.05	0.00	0.00
iLAM	Gla	2.20	0.48	0.13	0.00	0.00
iLAM	Acp5	2.12	0.54	0.17	0.00	0.00
iLAM	Plaur	2.11	0.68	0.25	0.00	0.00
iLAM	Gpr137b-ps	2.09	0.38	0.08	0.00	0.00
iLAM	Arl5c	2.09	0.46	0.11	0.00	0.00
iLAM	Il7r	2.04	0.25	0.09	0.00	0.00
iLAM	Ms4a4c	2.03	0.48	0.13	0.00	0.00
iLAM	Tgm2	2.03	0.59	0.16	0.00	0.00
iLAM	Abcg1	2.01	0.82	0.23	0.00	0.00
iLAM	Slamf8	1.98	0.40	0.12	0.00	0.00
iLAM	Gpnmb	1.96	0.49	0.11	0.00	0.00
iLAM	Fabp5	1.95	0.62	0.26	0.00	0.00
iLAM	Stx3	1.89	0.26	0.08	0.00	0.00
iLAM	2310022A10Rik	1.88	0.36	0.09	0.00	0.00
IFNIC Macrophages	Ifit2	8.55	0.67	0.01	0.00	0.00
IFNIC Macrophages	Ifit1bl1	8.06	0.27	0.00	0.00	0.00
IFNIC Macrophages	Rsad2	7.49	0.67	0.03	0.00	0.00
IFNIC Macrophages	Ifit3	7.22	1.00	0.03	0.00	0.00
IFNIC Macrophages	Ifit3b	6.88	0.73	0.01	0.00	0.00

cluster	gene	avg_log2FC	pct.1	pct.2	p_val	p_val_adj
IFNIC Macrophages	Ifit1	6.69	0.73	0.03	0.00	0.00
IFNIC Macrophages	Gbp5	6.26	0.33	0.01	0.00	0.00
IFNIC Macrophages	Cmpk2	6.25	0.47	0.02	0.00	0.00
IFNIC Macrophages	Gbp2	5.96	0.80	0.05	0.00	0.00
IFNIC Macrophages	Mx1	5.95	0.53	0.02	0.00	0.00
IFNIC Macrophages	Serpina3g	5.94	0.27	0.00	0.00	0.00
IFNIC Macrophages	Tgtp2	5.71	0.27	0.01	0.00	0.00
IFNIC Macrophages	Ifi205	5.05	0.53	0.03	0.00	0.00
IFNIC Macrophages	Isg15	5.02	0.87	0.14	0.00	0.00
IFNIC Macrophages	Usp18	5.01	0.60	0.05	0.00	0.00
IFNIC Macrophages	Pydc3	4.99	0.40	0.01	0.00	0.00
IFNIC Macrophages	Iigp1	4.93	0.47	0.02	0.00	0.00
IFNIC Macrophages	Oasl1	4.84	0.60	0.03	0.00	0.00
IFNIC Macrophages	Phf11a	4.54	0.33	0.04	0.00	0.00
IFNIC Macrophages	Ifi47	4.45	0.73	0.10	0.00	0.00
IFNIC Macrophages	Gm8953	4.44	0.27	0.01	0.00	0.00
IFNIC Macrophages	Gbp4	4.42	0.27	0.01	0.00	0.00
IFNIC Macrophages	Isg20	4.36	0.27	0.02	0.00	0.00
IFNIC Macrophages	Irgm2	4.33	0.53	0.03	0.00	0.00
IFNIC Macrophages	Mov10	4.27	0.27	0.02	0.00	0.00
IFNIC Macrophages	Ifih1	4.25	0.60	0.07	0.00	0.00
IFNIC Macrophages	Pydc4	4.03	0.67	0.06	0.00	0.00
IFNIC Macrophages	Herc6	4.03	0.53	0.05	0.00	0.00
IFNIC Macrophages	Gbp7	4.01	0.60	0.09	0.00	0.00
IFNIC Macrophages	Irgm1	3.95	0.53	0.09	0.00	0.00

Table S2: Top 30 differentially expressed genes for each annotated cell type in mouse, sorted on descending fold change.

6.5.2 Human macrophage gene markers

cluster	gene	avg_log2FC	pct.1	pct.2	p_val	p_val_adj
Inflammatory macrophages	FCGBP	2.46	0.35	0.12	0.00	0.00
Inflammatory macrophages	CX3CR1	1.87	0.34	0.12	0.00	0.00
Inflammatory macrophages	PLD4	1.86	0.26	0.09	0.00	0.00
Inflammatory macrophages	C3	1.42	0.38	0.21	0.00	0.00
Inflammatory macrophages	HTRA1	1.39	0.40	0.21	0.00	0.00
Inflammatory macrophages	USP53	1.38	0.41	0.21	0.00	0.00
Inflammatory macrophages	ALOX5AP	1.29	0.58	0.34	0.00	0.00
Inflammatory macrophages	VMO1	1.25	0.57	0.35	0.00	0.00
Inflammatory macrophages	BHLHE41	1.21	0.32	0.17	0.00	0.00
Inflammatory macrophages	IGSF21	1.15	0.36	0.18	0.00	0.00
Inflammatory macrophages	GAPT	1.13	0.26	0.13	0.00	0.00

cluster	gene	avg_log2FC	pct.1	pct.2	p_val	p_val_adj
Inflammatory macrophages	ETV5	1.00	0.33	0.21	0.00	0.00
Inflammatory macrophages	P2RY13	1.00	0.28	0.16	0.00	0.00
Inflammatory macrophages	SAP30	0.97	0.48	0.34	0.00	0.00
Inflammatory macrophages	LTC4S	0.97	0.46	0.28	0.00	0.00
Inflammatory macrophages	PDGFB	0.93	0.26	0.15	0.00	0.00
Inflammatory macrophages	SYNE1	0.89	0.30	0.21	0.00	0.00
Inflammatory macrophages	FILIP1L	0.89	0.36	0.23	0.00	0.00
Inflammatory macrophages	YWHAH	0.87	0.70	0.55	0.00	0.00
Inflammatory macrophages	LPAR6	0.85	0.49	0.31	0.00	0.00
Inflammatory macrophages	TM6SF1	0.85	0.34	0.22	0.00	0.00
Inflammatory macrophages	LPCAT2	0.85	0.43	0.31	0.00	0.00
Inflammatory macrophages	ITGA4	0.84	0.54	0.38	0.00	0.00
Inflammatory macrophages	OGFRL1	0.81	0.63	0.48	0.00	0.00
Inflammatory macrophages	MNDA	0.79	0.74	0.57	0.00	0.00
Inflammatory macrophages	HLA-DPA1	0.78	0.97	0.87	0.00	0.00
Inflammatory macrophages	HLA-DPB1	0.77	0.97	0.87	0.00	0.00
Inflammatory macrophages	HERPUD1	0.77	0.74	0.60	0.00	0.00
Inflammatory macrophages	HACD4	0.77	0.33	0.21	0.00	0.00
Inflammatory macrophages	RGS1	0.77	0.67	0.51	0.00	0.00
Resident-like macrophages	CD163L1	3.27	0.27	0.04	0.00	0.00
Resident-like macrophages	SLC40A1	2.78	0.67	0.22	0.00	0.00
Resident-like macrophages	IGF1	2.71	0.50	0.12	0.00	0.00
Resident-like macrophages	SELENOP	2.59	0.92	0.39	0.00	0.00
Resident-like macrophages	IL2RA	2.58	0.31	0.08	0.00	0.00
Resident-like macrophages	CXCL12	2.30	0.49	0.14	0.00	0.00
Resident-like macrophages	F13A1	2.24	0.77	0.40	0.00	0.00
Resident-like macrophages	ENPP2	2.20	0.31	0.10	0.00	0.00
Resident-like macrophages	LILRB5	2.04	0.26	0.08	0.00	0.00
Resident-like macrophages	PDGFC	2.01	0.27	0.10	0.00	0.00
Resident-like macrophages	PK4	1.96	0.49	0.23	0.00	0.00
Resident-like macrophages	CR1	1.95	0.35	0.13	0.00	0.00
Resident-like macrophages	FOLR2	1.92	0.80	0.34	0.00	0.00
Resident-like macrophages	WWP1	1.90	0.32	0.14	0.00	0.00
Resident-like macrophages	ME1	1.79	0.35	0.14	0.00	0.00
Resident-like macrophages	MS4A4A	1.74	0.80	0.48	0.00	0.00
Resident-like macrophages	PLTP	1.73	0.86	0.55	0.00	0.00
Resident-like macrophages	CMKLR1	1.68	0.31	0.12	0.00	0.00
Resident-like macrophages	MRC1	1.60	0.68	0.37	0.00	0.00
Resident-like macrophages	GIMAP5	1.59	0.28	0.12	0.00	0.00
Resident-like macrophages	LGMN	1.55	0.75	0.44	0.00	0.00
Resident-like macrophages	MAF	1.55	0.70	0.41	0.00	0.00
Resident-like macrophages	SIGLEC1	1.54	0.34	0.15	0.00	0.00
Resident-like macrophages	SLCO2B1	1.53	0.74	0.44	0.00	0.00

cluster	gene	avg_log2FC	pct.1	pct.2	p_val	p_val_adj
Resident-like macrophages	GAS6	1.53	0.58	0.31	0.00	0.00
Resident-like macrophages	C2	1.51	0.41	0.20	0.00	0.00
Resident-like macrophages	C1QA	1.50	0.96	0.66	0.00	0.00
Resident-like macrophages	STAB1	1.48	0.77	0.53	0.00	0.00
Resident-like macrophages	SDC3	1.47	0.52	0.26	0.00	0.00
Resident-like macrophages	RGL1	1.47	0.34	0.17	0.00	0.00
iLAMs	S100A12	3.47	0.28	0.04	0.00	0.00
iLAMs	G0S2	3.26	0.51	0.12	0.00	0.00
iLAMs	AQP9	2.89	0.49	0.10	0.00	0.00
iLAMs	CCL20	2.85	0.26	0.06	0.00	0.00
iLAMs	VCAN	2.80	0.71	0.21	0.00	0.00
iLAMs	TIMP1	2.76	0.86	0.58	0.00	0.00
iLAMs	EREG	2.71	0.46	0.09	0.00	0.00
iLAMs	S100A8	2.57	0.67	0.23	0.00	0.00
iLAMs	PHLDA1	2.49	0.46	0.14	0.00	0.00
iLAMs	CD300E	2.39	0.39	0.09	0.00	0.00
iLAMs	LUCAT1	2.25	0.50	0.16	0.00	0.00
iLAMs	MMP19	2.24	0.31	0.11	0.00	0.00
iLAMs	AC015912.3	2.23	0.29	0.10	0.00	0.00
iLAMs	C15orf48	2.22	0.62	0.26	0.00	0.00
iLAMs	PPIF	2.20	0.52	0.22	0.00	0.00
iLAMs	EHD1	2.17	0.29	0.10	0.00	0.00
iLAMs	IL1RN	2.17	0.47	0.15	0.00	0.00
iLAMs	ANPEP	2.13	0.46	0.15	0.00	0.00
iLAMs	SLC25A37	2.09	0.46	0.20	0.00	0.00
iLAMs	FBP1	2.00	0.58	0.30	0.00	0.00
iLAMs	FCN1	1.96	0.64	0.16	0.00	0.00
iLAMs	MAP3K20	1.93	0.27	0.09	0.00	0.00
iLAMs	SMIM25	1.87	0.54	0.20	0.00	0.00
iLAMs	S100A9	1.85	0.83	0.57	0.00	0.00
iLAMs	SLC2A3	1.84	0.58	0.30	0.00	0.00
iLAMs	MT1X	1.81	0.34	0.25	0.00	0.00
iLAMs	PLIN2	1.79	0.70	0.50	0.00	0.00
iLAMs	AREG	1.79	0.41	0.15	0.00	0.00
iLAMs	SLC11A1	1.76	0.66	0.34	0.00	0.00
iLAMs	MARCO	1.75	0.45	0.13	0.00	0.00
LAMs	MMP9	4.50	0.35	0.07	0.00	0.00
LAMs	CCL18	3.48	0.33	0.06	0.00	0.00
LAMs	APOC1	3.13	0.90	0.39	0.00	0.00
LAMs	NUPR1	2.54	0.54	0.18	0.00	0.00
LAMs	SPARC	2.43	0.26	0.06	0.00	0.00
LAMs	APOC2	2.34	0.26	0.10	0.00	0.00
LAMs	RAB13	2.31	0.56	0.34	0.00	0.00

cluster	gene	avg_log2FC	pct.1	pct.2	p_val	p_val_adj
LAMs	OTOA	2.29	0.40	0.13	0.00	0.00
LAMs	CSTB	2.22	0.94	0.81	0.00	0.00
LAMs	APOE	2.20	0.84	0.41	0.00	0.00
LAMs	SPP1	2.09	0.73	0.45	0.00	0.00
LAMs	MATK	2.07	0.26	0.07	0.00	0.00
LAMs	ACP5	1.98	0.53	0.29	0.00	0.00
LAMs	SCD	1.97	0.40	0.16	0.00	0.00
LAMs	FTL	1.92	1.00	1.00	0.00	0.00
LAMs	CYP27A1	1.90	0.33	0.12	0.00	0.00
LAMs	GCHFR	1.81	0.30	0.13	0.00	0.00
LAMs	GPNUMB	1.74	0.71	0.43	0.00	0.00
LAMs	VAT1	1.56	0.44	0.20	0.00	0.00
LAMs	FN1	1.55	0.43	0.27	0.00	0.00
LAMs	LGALS3	1.52	0.82	0.68	0.00	0.00
LAMs	CYSTM1	1.46	0.50	0.27	0.00	0.00
LAMs	CYB5A	1.36	0.36	0.20	0.00	0.00
LAMs	CAPG	1.35	0.80	0.68	0.00	0.00
LAMs	TXN	1.35	0.76	0.61	0.00	0.00
LAMs	TNS1	1.34	0.25	0.11	0.00	0.00
LAMs	MCRIP2	1.32	0.33	0.17	0.00	0.00
LAMs	CD109	1.32	0.27	0.12	0.00	0.00
LAMs	LGALS1	1.30	0.96	0.85	0.00	0.00
LAMs	PLD3	1.29	0.68	0.56	0.00	0.00
IFNIC macrophages	CXCL10	6.23	0.32	0.02	0.00	0.00
IFNIC macrophages	RSAD2	4.83	0.44	0.03	0.00	0.00
IFNIC macrophages	IFIT1	4.53	0.60	0.08	0.00	0.00
IFNIC macrophages	ISG15	3.89	0.88	0.34	0.00	0.00
IFNIC macrophages	HERC5	3.78	0.50	0.08	0.00	0.00
IFNIC macrophages	CMPK2	3.75	0.41	0.04	0.00	0.00
IFNIC macrophages	HELZ2	3.71	0.27	0.03	0.00	0.00
IFNIC macrophages	MX1	3.69	0.82	0.18	0.00	0.00
IFNIC macrophages	OAS3	3.41	0.50	0.08	0.00	0.00
IFNIC macrophages	GBP1	3.36	0.50	0.13	0.00	0.00
IFNIC macrophages	IFIT3	3.32	0.63	0.11	0.00	0.00
IFNIC macrophages	IFI44L	3.27	0.75	0.15	0.00	0.00
IFNIC macrophages	SAMD9L	3.10	0.61	0.14	0.00	0.00
IFNIC macrophages	OAS2	3.04	0.55	0.10	0.00	0.00
IFNIC macrophages	DDX58	3.02	0.38	0.07	0.00	0.00
IFNIC macrophages	ISG20	3.01	0.35	0.08	0.00	0.00
IFNIC macrophages	CCL8	2.91	0.28	0.08	0.00	0.00
IFNIC macrophages	OASL	2.90	0.36	0.06	0.00	0.00
IFNIC macrophages	MX2	2.85	0.73	0.22	0.00	0.00
IFNIC macrophages	TNFSF10	2.84	0.67	0.24	0.00	0.00

cluster	gene	avg_log2FC	pct.1	pct.2	p_val	p_val_adj
IFNIC macrophages	GBP5	2.78	0.26	0.07	0.00	0.00
IFNIC macrophages	IFIT2	2.77	0.59	0.16	0.00	0.00
IFNIC macrophages	LY6E	2.65	0.82	0.32	0.00	0.00
IFNIC macrophages	XAF1	2.63	0.70	0.20	0.00	0.00
IFNIC macrophages	STAT1	2.53	0.75	0.25	0.00	0.00
IFNIC macrophages	IFI6	2.52	0.85	0.36	0.00	0.00
IFNIC macrophages	GBP4	2.39	0.36	0.11	0.00	0.00
IFNIC macrophages	EPSTI1	2.38	0.74	0.25	0.00	0.00
IFNIC macrophages	IFIH1	2.38	0.41	0.13	0.00	0.00
IFNIC macrophages	OAS1	2.37	0.62	0.21	0.00	0.00
Proliferating macrophages	MKI67	6.75	0.36	0.01	0.00	0.00
Proliferating macrophages	FAM111B	6.49	0.29	0.00	0.00	0.00
Proliferating macrophages	PCLAF	6.34	0.52	0.01	0.00	0.00
Proliferating macrophages	CENPM	6.02	0.33	0.01	0.00	0.00
Proliferating macrophages	TYMS	5.77	0.44	0.01	0.00	0.00
Proliferating macrophages	UHRF1	5.74	0.38	0.01	0.00	0.00
Proliferating macrophages	CDCA7	5.59	0.36	0.01	0.00	0.00
Proliferating macrophages	E2F1	5.46	0.28	0.01	0.00	0.00
Proliferating macrophages	GINS2	5.29	0.44	0.01	0.00	0.00
Proliferating macrophages	BIRC5	5.11	0.29	0.01	0.00	0.00
Proliferating macrophages	TK1	5.07	0.51	0.02	0.00	0.00
Proliferating macrophages	CDT1	5.06	0.35	0.01	0.00	0.00
Proliferating macrophages	CLSPN	4.92	0.52	0.02	0.00	0.00
Proliferating macrophages	MAD2L1	4.41	0.34	0.01	0.00	0.00
Proliferating macrophages	CDKN3	4.31	0.32	0.01	0.00	0.00
Proliferating macrophages	HELLS	4.29	0.48	0.02	0.00	0.00
Proliferating macrophages	CENPK	4.28	0.47	0.02	0.00	0.00
Proliferating macrophages	MCM2	4.20	0.35	0.02	0.00	0.00
Proliferating macrophages	TCF19	4.10	0.30	0.02	0.00	0.00
Proliferating macrophages	ZWINT	4.10	0.41	0.02	0.00	0.00
Proliferating macrophages	NUSAP1	4.08	0.28	0.02	0.00	0.00
Proliferating macrophages	C12orf75	3.88	0.38	0.02	0.00	0.00
Proliferating macrophages	CENPF	3.86	0.27	0.02	0.00	0.00
Proliferating macrophages	UBE2T	3.82	0.27	0.02	0.00	0.00
Proliferating macrophages	ORC6	3.75	0.38	0.02	0.00	0.00
Proliferating macrophages	CD1C	3.74	0.28	0.03	0.00	0.00
Proliferating macrophages	MCM4	3.69	0.51	0.04	0.00	0.00
Proliferating macrophages	CHEK1	3.66	0.30	0.02	0.00	0.00
Proliferating macrophages	ASF1B	3.65	0.27	0.02	0.00	0.00
Proliferating macrophages	FEN1	3.54	0.43	0.03	0.00	0.00

Table S3: Top 30 differentially expressed genes for each annotated cell type in human, sorted on descending fold change.
